

Asymptotic Normality and Efficiency of Maximum Likelihood in Probabilistic PCA on the Quotient Manifold $\Theta/O(q)$

Working draft v3: fully detailed proofs and statistical consequences

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Abstract

This draft carries out, with detailed proofs, the program of answering the open problem in Datta’s thesis (§1.8, §4.1): formulating and proving asymptotic normality and efficiency of the maximum likelihood estimator (MLE) of probabilistic PCA on the quotient manifold $\mathcal{Q} = \Theta/O(q)$ built in §4.1.1 of the thesis, in the intrinsic framework of Sun, Lin and Liu (arXiv:2510.13703). Relative to v2, every proof has been expanded: the geometry of $\mathcal{Q}$ is developed from first principles (horizontal lifts, a local distance identity, a self-contained proof that projected horizontal lines are geodesics, a chord-equality argument for uniqueness of minimizing geodesics, an explicit parallel-transport ODE, and an O’Neill curvature bound with all derivatives computed); the differentiability in quadratic mean, asymptotic linearity, and local regularity of the MLE are proved with full verification of every hypothesis of the classical theorems invoked; the Fisher information operator and its inverse are computed in closed form in interpretable block coordinates, with the score coordinates derived explicitly and validated by moment identities; the efficient influence function of $\hat{\sigma}_n^2$ is derived twice — once from G^{-1} s and once, independently, from eigenvalue perturbation theory via a Davis–Kahan bound; and an almost-sure well-posedness lemma for the Procrustes-aligned estimation error is proved by a real-analyticity argument. A new section analyzes the *consequences* of the result: the geometric picture of a sequence of random tangent vectors converging to a Gaussian, and a concrete guide for the applied statistician — standard errors, Wald and likelihood-ratio tests with the correct (reduced) degrees of freedom, confidence regions as exponential images of ellipsoids, aligned bootstrap, and diagnostics. No eigengap condition on W_0 is required anywhere.

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1 Introduction

1.1 The model and the obstruction

PPCA [14] posits $x = Wz + \varepsilon$ with latent $z \sim \mathcal{N}(0, I_q)$, noise $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_p)$ independent of z , and $q < p$, so that marginally

$$x \sim \mathcal{N}(0, \Sigma(\theta)), \quad \Sigma(\theta) := WW^\top + \sigma^2 I_p, \quad \theta = (W, \sigma^2).$$

Since $\Sigma(WR, \sigma^2) = \Sigma(W, \sigma^2)$ for every $R \in O(q)$, the parameter is identified only up to the right action of the orthogonal group, and the naive claim “ $\sqrt{n}(\widehat{W}_n - W_0)$ is asymptotically normal” is ill-posed: $\widehat{W}_n$ need not converge to W_0 ; only its orbit does.

1.2 Provenance and inputs

Datta’s thesis, Chapter 1 (published with Chakrabarty [5]): consistency of the MLE $\hat{\theta}_n = (\widehat{W}_n, \widehat{\sigma}_n^2)$ in the quotient Θ/C , $C = \{\theta : \Sigma(\theta) = \Sigma(\theta_0)\}$, in the Redner-style metric $\bar{d}$ [12]; strong consistency for i.i.d. data, weak consistency under a weak law of large numbers. §1.8 leaves rates and limit distributions open.

Datta’s thesis, §4.1–4.1.1: the open problem is restated — convergence rates, or a stronger statement such as a CLT for the MLE inside the quotient — and the geometric prerequisite is supplied: C is the $O(q)$ -orbit of θ_0 (Theorem 13); Θ/C is canonically the orbit space $\Theta/O(q)$, a smooth manifold of dimension $pq + 1 - q(q - 1)/2$ (Proposition 9); the Euclidean product metric descends to a quotient Riemannian metric (Corollary 4).

Sun, Lin and Liu [13]: the asymptotic efficiency theory à la Le Cam for parameter *manifolds*: intrinsic estimation error, differentiability in quadratic mean (DQM), local asymptotic normality (LAN), regular estimators, influence operators, convolution and local asymptotic minimax (LAM) theorems. Their Table 1 is the dictionary this draft instantiates at $\mathcal{Q}$.

1.3 Contributions of this draft

- (i) A self-contained geometric toolkit for $\mathcal{Q}$ (Section 2): projections via a Lyapunov equation; existence and length-preservation of horizontal lifts; a local identity between quotient distance and Euclidean orbit distance; a from-scratch proof that projected horizontal lines are the geodesics of $\mathcal{Q}$; explicit injectivity and uniform normal-neighborhood radii; the logarithm as a Procrustes alignment, with uniqueness of minimizing geodesics by a chord-equality argument; the parallel-transport ODE; the O’Neill tensor computed exactly and a curvature bound.
- (ii) Definitions reproducing [13] verbatim at $\Psi = \mathcal{Q}$, $\psi = \text{id}$ (Section 3), together with a new almost-sure well-posedness lemma for the aligned estimation error, proved by a Riesz-projector real-analyticity argument (Lemma 3.3).
- (iii) A complete proof of DQM and LAN of the quotient model (Theorem 5.2), exploiting the fact that $\text{Exp}_{[\theta]}$ is linear in the horizontal lift, and full verification of every hypothesis of van der Vaart’s Lemma 7.6 and Theorem 5.23 [15] (Section 6).
- (iv) Closed-form Fisher calculus (Section 4): explicit score coordinates, explicit G and G^{-1} in interpretable blocks, moment-identity verification, the efficient influence function $\text{IF}_{\sigma^2}(x) = (\|U_{\perp}^{\top} x\|^2 - (p - q)\sigma_0^2)/(p - q)$ derived twice (algebraically and via Davis–Kahan eigenvalue perturbation), and three cross-checks against Anderson’s classical PCA asymptotics — all without any eigengap assumption (Theorem 4.2).
- (v) Fully detailed proofs of asymptotic linearity, local regularity (via an elementary transport-comparison lemma with explicit smoothness), and efficiency by the convolution and LAM theorems of [13] (Section 6); a covariance-level CLT with a blockwise comparison to the sample covariance and an efficiency-dominance identity (Corollaries 5.5, 5.6, Proposition 5.7).
- (vi) A new section on *consequences* (Section 7): what asymptotic normality of tangent vectors means geometrically, and how an applied statistician turns the theorem into standard errors, tests, confidence regions, and diagnostics.

1.4 Notation

$\text{Sym}(m)$, $\text{Skew}(m)$: symmetric / skew-symmetric $m \times m$ matrices; $O(q)$: orthogonal group; $\text{sym}(M) = \frac{1}{2}(M + M^{\top})$, $\text{skew}(M) = \frac{1}{2}(M - M^{\top})$; $\|\cdot\|_F$, $\|\cdot\|_2$: Frobenius and spectral norms; $\sigma_{\max}$, $\sigma_{\min}$:

extreme singular values; $\lambda_j(\cdot)$: ordered eigenvalues. For $S \succ 0$, $\mathbb{L}_S[X] := SX + XS$ (Lemma B.1). $\theta_0 = (W_0, \sigma_0^2)$, $\Sigma_0 = \Sigma(\theta_0)$; $[\theta]$ is the class of θ ; $\text{Orb}(\theta)$ its orbit; $\rightsquigarrow_P$ is weak convergence under P . A thin SVD $W_0 = U\Lambda V^\top$ is fixed, with $U \in \mathbb{R}^{p \times q}$ orthonormal columns, $U_\perp$ completing U , $V \in O(q)$, and $\Lambda = \text{diag}(\ell_1, \dots, \ell_q)$, $\ell_1 \geq \dots \geq \ell_q > 0$. Set

$$D := \Lambda^2 + \sigma_0^2 I_q = \text{diag}(\lambda_1, \dots, \lambda_q), \quad \lambda_i = \ell_i^2 + \sigma_0^2,$$

so $\Sigma_0 = UDU^\top + \sigma_0^2 U_\perp U_\perp^\top$: the λ_i are the leading eigenvalues of Σ_0 and σ_0^2 its trailing eigenvalue, of multiplicity $p - q$ and separated from λ_q by the *population gap* $\lambda_q - \sigma_0^2 = \ell_q^2 > 0$, which exists without any assumption.

2 Geometry of the quotient parameter manifold

2.1 Total space, group action, quotient manifold

$$\Theta := \{(W, \sigma^2) \in \mathbb{R}^{p \times q} \times \mathbb{R}_{>0} : \text{rank}(W) = q\}, \quad (W, \sigma^2) \cdot R := (WR, \sigma^2), \quad R \in O(q).$$

The action is smooth; it is *free* because $WR = W$ with W of full column rank forces $R = I_q$; it is *proper* because $O(q)$ is compact. By the quotient manifold theorem ([2, Ch. 9]; Datta, Proposition 9),

$$\mathcal{Q} := \Theta / O(q), \quad \dim \mathcal{Q} = \dim \Theta - \dim O(q) = pq + 1 - \frac{q(q-1)}{2},$$

is a smooth manifold and $\pi : \Theta \rightarrow \mathcal{Q}$ a smooth submersion whose fibers are the orbits. Datta's Theorem 13 identifies the fiber over $[\theta_0]$ with the set C of Chapter 1, so Θ/C and $\mathcal{Q}$ describe the same statistical object. (Chapter 1 of the thesis works on $\mathbb{R}^{p \times q} \times [\epsilon_*, \infty)$ without a rank restriction; the restriction here is harmless because the MLE has rank q and $\liminf_n \hat{\sigma}_n^2 > 0$, as noted in §1.3 of the thesis, and $\text{rank}(W_0) = q$ is assumed.)

2.2 Metric, vertical and horizontal spaces, projections

Endow Θ with the Euclidean product metric $\langle (A, a), (B, b) \rangle = \text{tr}(A^\top B) + ab$; it is $O(q)$ -invariant because $\text{tr}((AR)^\top BR) = \text{tr}(R^\top A^\top BR) = \text{tr}(A^\top B)$ (Datta, Corollary 4). At $\theta = (W, \sigma^2)$, the orbit $R \mapsto (WR, \sigma^2)$ is an embedded submanifold; differentiating $t \mapsto (We^{t\Omega}, \sigma^2)$ at $t = 0$ shows its tangent space — the *vertical space* — is

$$\mathcal{V}_\theta = \{(W\Omega, 0) : \Omega \in \text{Skew}(q)\}, \quad \dim \mathcal{V}_\theta = \frac{q(q-1)}{2}, \quad (1)$$

the dimension count using that $\Omega \mapsto W\Omega$ is injective for W of full column rank. The *horizontal space* is its orthogonal complement: $(A, a) \perp (W\Omega, 0)$ for all skew Ω iff $\text{tr}(\Omega^\top W^\top A) = 0$ for all skew Ω iff the skew part of $W^\top A$ vanishes, i.e.

$$\mathcal{H}_\theta = \{(A, a) : W^\top A = A^\top W\}, \quad \dim \mathcal{H}_\theta = pq + 1 - \frac{q(q-1)}{2}. \quad (2)$$

The quotient metric is by definition the one making $d\pi_\theta : \mathcal{H}_\theta \rightarrow T_{[\theta]} \mathcal{Q}$ a linear isometry; we use this isometry throughout to identify $T_{[\theta]} \mathcal{Q} \cong \mathcal{H}_\theta$, and $\bar{h}$ denotes the horizontal lift of $h \in T_{[\theta]} \mathcal{Q}$. For an arbitrary tangent vector $\xi \in T_\theta \Theta$, $\|d\pi_\theta \xi\| = \|P_\theta^\mathcal{H} \xi\| \leq \|\xi\|$, with equality iff ξ is horizontal — the fact that *projection does not increase length*, used repeatedly below.

Lemma 2.1 (Projections). *For $(A, a) \in T_\theta \Theta$,*

$$P_\theta^\mathcal{V}(A, a) = (W\Omega_A, 0), \quad \mathbb{L}_{W^\top W}[\Omega_A] = W^\top A - A^\top W,$$

with $\Omega_A \in \text{Skew}(q)$ unique, and $P_\theta^\mathcal{H} = \text{id} - P_\theta^\mathcal{V}$.

Proof. We seek the skew Ω minimizing $\|A - W\Omega\|_F^2$, equivalently making $(A - W\Omega, a) \perp \mathcal{V}_\theta$: $\text{tr}(\Xi^\top W^\top (A - W\Omega)) = 0$ for all $\Xi \in \text{Skew}(q)$, i.e. $\text{skew}(W^\top A - W^\top W\Omega) = 0$. Since $(W^\top W\Omega)^\top = \Omega^\top W^\top W = -\Omega W^\top W$, we have $2\text{skew}(W^\top W\Omega) = W^\top W\Omega + \Omega W^\top W = \mathbb{L}_{W^\top W}[\Omega]$, so the condition reads $\mathbb{L}_{W^\top W}[\Omega] = W^\top A - A^\top W$. The right-hand side is skew; by Lemma B.1 the Lyapunov operator with $W^\top W \succ 0$ is invertible and maps $\text{Skew}(q)$ bijectively onto itself, so a unique skew solution Ω_A exists. $\square$

2.3 Horizontal lifts of curves

Lemma 2.2 (Horizontal lifts). *Let $c : [0, 1] \rightarrow \mathcal{Q}$ be piecewise C^1 with $c(0) = [\theta^*]$, and let $L := \text{length}(c)$. If the Euclidean ball $\{x : \|x - \theta^*\| \leq L\}$ is contained in Θ , then there is a unique piecewise- C^1 curve $\bar{c} : [0, 1] \rightarrow \Theta$ with $\pi \circ \bar{c} = c$, $\bar{c}(0) = \theta^*$, and $\dot{\bar{c}}(t) \in \mathcal{H}_{\bar{c}(t)}$ for all t ; moreover $\text{length}(\bar{c}) = \text{length}(c)$ and $\|\bar{c}(t) - \theta^*\| \leq \text{length}(c|_{[0,t]})$.*

Proof. Because π is a submersion with the smooth horizontal distribution (2) on which $d\pi$ is a linear isometry, the lift is the solution of the (piecewise-continuous in t , smooth in x) ODE

$$\dot{x}(t) = (d\pi_{x(t)}|_{\mathcal{H}_{x(t)}})^{-1} \dot{c}(t), \quad x(0) = \theta^*,$$

which is well posed by Picard–Lindelöf in each local trivialization of π (see [2, Ch. 9]); along any solution, $\frac{d}{dt}\pi(x(t)) = d\pi \dot{x} = \dot{c}$, so $\pi \circ x = c$, and $\|\dot{x}(t)\| = \|\dot{c}(t)\|$ by the isometry, giving both the length identity and the a priori bound $\|x(t) - \theta^*\| \leq \int_0^t \|\dot{x}\| \leq \text{length}(c|_{[0,t]}) \leq L$; the bound keeps $x(t)$ inside the stated Euclidean ball, hence inside Θ , so the solution extends to $[0, 1]$. Uniqueness is ODE uniqueness. $\square$

2.4 The local distance identity

For $\theta^* = (W^*, \sigma_*^2) \in \Theta$ set $\rho_2(\theta^*) := \frac{1}{2} \min\{\sigma_{\min}(W^*), \sigma_*^2\}$. Note that $\text{Orb}(\theta')$ is compact (continuous image of $\text{O}(q)$), so $d(\theta^*, \text{Orb}(\theta')) := \min_{R \in \text{O}(q)} \|(W'R, \sigma'^2) - \theta^*\|$ is attained.

Lemma 2.3 (Local distance identity). *Let $\theta^* \in \Theta$ and $\theta' \in \Theta$ with $\delta := d(\theta^*, \text{Orb}(\theta')) < \rho_2(\theta^*)$. Then*

$$d_{\mathcal{Q}}([\theta^*], [\theta']) = \delta.$$

Proof. Upper bound. Let $y^* \in \text{Orb}(\theta')$ attain the minimum. Every point on the segment $[\theta^*, y^*]$ lies within δ of θ^* , hence has $\sigma_{\min}(W) \geq \sigma_{\min}(W^*) - \delta > 0$ (Weyl) and $\sigma^2 \geq \sigma_*^2 - \delta > 0$, so the segment lies in Θ . Projecting it by π produces a path in $\mathcal{Q}$ from $[\theta^*]$ to $[\theta']$ of length $\leq \delta$ (projection does not increase length, §2.2); hence $d_{\mathcal{Q}} \leq \delta$.

Lower bound. Let c be any piecewise- C^1 path in $\mathcal{Q}$ from $[\theta^*]$ to $[\theta']$ with $\text{length}(c) < \delta + \varepsilon \leq 2\rho_2(\theta^*) \leq \min\{\sigma_{\min}(W^*), \sigma_*^2\}$, where $\varepsilon \in (0, \rho_2)$. By Lemma 2.2 (the Euclidean ball of radius $\text{length}(c)$ around θ^* lies in Θ by the same Weyl argument), c lifts horizontally from θ^* to $\bar{c}$ with $\text{length}(\bar{c}) = \text{length}(c)$ and $\bar{c}(1) \in \pi^{-1}([\theta']) = \text{Orb}(\theta')$. Then $\text{length}(c) = \text{length}(\bar{c}) \geq \|\bar{c}(1) - \theta^*\| \geq \delta$. Taking the infimum over such c (paths of length $\geq \delta + \varepsilon$ satisfy the bound trivially) gives $d_{\mathcal{Q}} \geq \delta$. $\square$

2.5 Horizontal lines, geodesics, and the exponential map

Lemma 2.4 (Horizontality and optimal alignment along a line). *Let $(A, a) \in \mathcal{H}_\theta$ and $W(t) := W + tA$, defined while the line stays in Θ . Then:*

- (a) $W(t)^\top A = W^\top A + tA^\top A$ is symmetric for every t , i.e. the line has horizontal velocity throughout;

(b) if in addition $W(t_1)^\top W(t_2) \succeq 0$ for $0 \leq t_1 < t_2$, then $R = I_q$ solves the Procrustes problem $\min_{R \in O(q)} \|W(t_2)R - W(t_1)\|_F$, so that

$$d(\theta(t_1), \text{Orb}(\theta(t_2))) = (t_2 - t_1) \|(A, a)\|, \quad \theta(t) := \theta + t(A, a).$$

Proof. (a) Both $W^\top A$ (horizontality at θ) and $A^\top A$ are symmetric. (b) $W(t_1)^\top W(t_2) = W(t_1)^\top W(t_1) + (t_2 - t_1)W(t_1)^\top A$ is symmetric by (a), and $\succeq 0$ by hypothesis; by the characterization in Lemma B.3, $R = I_q$ is a Procrustes minimizer, and the minimum value is $\|W(t_2) - W(t_1)\|_F^2 + (t_2 - t_1)^2 a^2 = (t_2 - t_1)^2 \|(A, a)\|^2$ (the σ^2 -part does not involve R). $\square$

Fix $\rho_1 \in (0, \rho_0]$, where $\rho_0 := \min\{\ell_q^2/\ell_1, \sigma_0^2\}$, small enough that

$$(\ell_1 + \rho_1)\rho_1 < (\ell_q - \rho_1)^2 \quad \text{and} \quad \rho_1 < \frac{1}{2} \min\{\ell_q - \rho_1, \sigma_0^2 - \rho_1\}; \quad (3)$$

such ρ_1 exists because as $\rho_1 \downarrow 0$ the left sides tend to 0 and the right sides to $\ell_q^2 > 0$ and $\frac{1}{2} \min\{\ell_q, \sigma_0^2\} > 0$.

Proposition 2.5 (Geodesics through $[\theta_0]$ and the exponential map). *For $h \in \mathcal{H}_{\theta_0}$ with $\|h\| < \rho_1$, the curve $\gamma_h(t) := [\theta_0 + t\bar{h}]$, $t \in [0, 1]$, is the unique constant-speed geodesic of $\mathcal{Q}$ with $\gamma_h(0) = [\theta_0]$ and $\dot{\gamma}_h(0) = h$; consequently*

$$\text{Exp}_{[\theta_0]}(h) = [\theta_0 + \bar{h}]. \quad (4)$$

Proof. Write $\bar{h} = (A, a)$, $\theta(t) = \theta_0 + t\bar{h}$, $W(t) = W_0 + tA$. For $t \in [0, 1]$, Weyl gives $\sigma_{\min}(W(t)) \geq \ell_q - \rho_1 > 0$ and $\sigma^2(t) \geq \sigma_0^2 - \rho_1 > 0$, so the line stays in Θ and the curve γ_h is well defined, with constant speed $\|d\pi \bar{h}\| = \|\bar{h}\| = \|h\|$ by Lemma 2.4(a) and the isometry of $d\pi$ on horizontal vectors.

Local minimality. For $0 \leq t_1 < t_2 \leq 1$: $W(t_1)^\top W(t_2) = W(t_1)^\top W(t_1) + (t_2 - t_1)W(t_1)^\top A$, where $\lambda_{\min}(W(t_1)^\top W(t_1)) \geq (\ell_q - \rho_1)^2$ and $\|(t_2 - t_1)W(t_1)^\top A\|_2 \leq (\ell_1 + \rho_1)\rho_1$, so by the first condition in (3) the sum is positive definite; Lemma 2.4(b) then gives $d(\theta(t_1), \text{Orb}(\theta(t_2))) = (t_2 - t_1)\|h\| \leq \rho_1$. The second condition in (3) guarantees $(t_2 - t_1)\|h\| < \rho_2(\theta(t_1))$ (indeed $\rho_2(\theta(t_1)) = \frac{1}{2} \min\{\sigma_{\min}(W(t_1)), \sigma^2(t_1)\} \geq \frac{1}{2} \min\{\ell_q - \rho_1, \sigma_0^2 - \rho_1\} > \rho_1$), so Lemma 2.3 applies at the base point $\theta(t_1)$:

$$d_{\mathcal{Q}}(\gamma_h(t_1), \gamma_h(t_2)) = (t_2 - t_1)\|h\| = \text{length}(\gamma_h|_{[t_1, t_2]}).$$

Thus γ_h is a constant-speed curve each of whose subsegments is minimizing; every such curve is a geodesic ([7, Ch. 6]). Uniqueness with the given initial conditions is the ODE uniqueness of geodesics ([7, Ch. 4]), and (4) is the definition of the exponential map evaluated on γ_h . $\square$

2.6 Injectivity and a uniform normal radius

Lemma 2.6 (Injectivity at θ_0). *The map $h \mapsto [\theta_0 + \bar{h}]$ is injective on $\{h \in \mathcal{H}_{\theta_0} : \|h\| < \rho_0\}$, and is a diffeomorphism onto its image.*

Proof. Suppose $[\theta_0 + \bar{h}_1] = [\theta_0 + \bar{h}_2]$ with $\bar{h}_i = (A_i, a_i)$, $\|h_i\| < \rho_0$. The σ^2 -components force $a_1 = a_2$, and $(W_0 + A_1)R = W_0 + A_2$ for some $R \in O(q)$. Multiplying by $W_0^\top$ gives $M_1 R = M_2$ with $M_i := W_0^\top W_0 + W_0^\top A_i$. Each M_i is symmetric (horizontalities of h_i) and positive definite: $\|W_0^\top A_i\|_2 \leq \ell_1 \|A_i\|_2 \leq \ell_1 \|h_i\| < \ell_1 \rho_0 \leq \ell_q^2 = \lambda_{\min}(W_0^\top W_0)$. Hence $M_1 = M_1 \cdot I_q$ and $M_1 = M_2 \cdot R^\top$ are two factorizations of the invertible matrix M_1 as (symmetric positive definite) $\times$ (orthogonal); by uniqueness of the polar decomposition (Lemma B.2), $R^\top = I_q$, so $A_1 = A_2$. Since $h \mapsto \theta_0 + \bar{h}$ is affine and π is a submersion whose differential restricted to $\mathcal{H}$ is an isomorphism, the map is a smooth injective immersion, hence a diffeomorphism onto its image by the inverse function theorem combined with injectivity. $\square$

Lemma 2.7 (Uniform normal radius on a ball). *There exists $r_0 > 0$, depending only on $(\ell_1, \ell_q, \sigma_0^2)$, with the following properties. Every $\theta^* \in \Theta$ with $d(\theta^*, \text{Orb}(\theta_0)) \leq 2r_0$ satisfies $\sigma_{\min}(W^*) \in [\ell_q - 2r_0, \ell_1 + 2r_0]$ and $\sigma_*^2 \geq \sigma_0^2 - 2r_0$, and the conclusions of Proposition 2.5, Lemma 2.6 and Proposition 2.8 below hold at the base point θ^* with a radius $\geq 8r_0$ in place of ρ_1, ρ_0, r_0 . In particular every point of the closed metric ball $\bar{B}([\theta_0], r_0)$ possesses a normal neighborhood containing the whole ball.*

Proof. All constants entering (3), Lemma 2.6 and Proposition 2.8 are continuous, strictly positive functions of $(\sigma_{\min}(W^*), \sigma_{\max}(W^*), \sigma_*^2)$, which for representatives within Euclidean distance $2r_0$ of $\text{Orb}(\theta_0)$ range over the compact set $[\ell_q - 2r_0, \ell_1 + 2r_0]^2 \times [\sigma_0^2 - 2r_0, \sigma_0^2 + 2r_0]$ by Weyl's inequality and the $O(q)$ -invariance of singular values. Hence their infimum over the ball is positive for r_0 small, and one may choose r_0 so small that this infimum exceeds $8r_0$; the last statement follows because any two points of the ball are at d_Q -distance $\leq 2r_0 < 8r_0$ from each other. $\square$

Throughout, r_0 is fixed as in Lemma 2.7 (further finite shrinkings, always depending only on $(\ell_1, \ell_q, \sigma_0^2)$, are noted where used).

2.7 Logarithm, distance, Procrustes alignment

Proposition 2.8 (Logarithm = Procrustes alignment). *Let $[\theta']$ satisfy $\delta := d(\theta_0, \text{Orb}(\theta')) < r_0$, and let $\hat{R} = \bar{U}\bar{V}^\top$ for any SVD $W'^\top W_0 = \bar{U}\bar{S}\bar{V}^\top$. Then:*

- (a) $W_0^\top W'$ is nonsingular and $\hat{R}$ is the unique Procrustes minimizer; the closest representative $y^* := (W'\hat{R}, \sigma'^2)$ of $\text{Orb}(\theta')$ to θ_0 is unique;
- (b) the aligned difference $\bar{h}^* := (W'\hat{R} - W_0, \sigma'^2 - \sigma_0^2)$ is horizontal at θ_0 , and $t \mapsto [\theta_0 + t\bar{h}^*]$ is a minimizing geodesic from $[\theta_0]$ to $[\theta']$ of length δ ;
- (c) it is the unique minimizing geodesic; consequently

$$\text{Exp}_{[\theta_0]}^{-1}([\theta']) = \bar{h}^* = (W'\hat{R} - W_0, \sigma'^2 - \sigma_0^2) \in \mathcal{H}_{\theta_0}, \quad d_Q([\theta_0], [\theta']) = \delta. \quad (5)$$

In particular $\text{Exp}_{[\theta_0]}$ maps $\{h : \|h\| < r_0\}$ diffeomorphically onto an open set containing the metric ball $B([\theta_0], r_0)$, which is therefore a normal neighborhood of $[\theta_0]$.

Proof. (a) By Lemma B.3, minimizers exist and any minimizer R satisfies $W_0^\top W'R \in \text{Sym}(q)$, $\succeq 0$. For such R , $\|W'R - W_0\|_F \leq \delta < r_0$, hence $W_0^\top W'R = W_0^\top W_0 + W_0^\top (W'R - W_0) \succ 0$ because $\|W_0^\top (W'R - W_0)\|_2 \leq \ell_1 r_0 < \ell_q^2$ (shrink r_0 if necessary). In particular $W_0^\top W'$ is nonsingular, and Lemma B.3 then gives uniqueness of the minimizer and the closed form $\hat{R} = \bar{U}\bar{V}^\top$; y^* is unique because the σ^2 -coordinate is fixed.

(b) Horizontality: $W_0^\top (W'\hat{R} - W_0) = W_0^\top W'\hat{R} - W_0^\top W_0 \in \text{Sym}(q)$ by (a). Since $\|\bar{h}^*\| = \delta < r_0 < \rho_1$ (Lemma 2.7), Proposition 2.5 shows $t \mapsto [\theta_0 + t\bar{h}^*]$ is a geodesic; its endpoint is $[y^*] = [\theta']$ and its length is $\|\bar{h}^*\| = \delta$, which equals $d_Q([\theta_0], [\theta'])$ by Lemma 2.3; so it is minimizing.

(c) Let c be any minimizing geodesic from $[\theta_0]$ to $[\theta']$, of length δ . By Lemma 2.2 it lifts horizontally from θ_0 to $\bar{c}$ with $\text{length}(\bar{c}) = \delta$ and $\bar{c}(1) \in \text{Orb}(\theta')$; then

$$\delta = \text{length}(\bar{c}) \geq \|\bar{c}(1) - \theta_0\| \geq d(\theta_0, \text{Orb}(\theta')) = \delta,$$

forcing equality throughout. Equality in the length–chord inequality forces $\bar{c}$ to be a monotone parametrization of the straight segment: with e the unit vector along $\bar{c}(1) - \theta_0$, $\text{length}(\bar{c}) = \int_0^1 \|\dot{\bar{c}}\| \geq \int_0^1 \langle \dot{\bar{c}}, e \rangle = \|\bar{c}(1) - \theta_0\|$, and equality holds iff $\dot{\bar{c}}(t) = \|\dot{\bar{c}}(t)\| e$ a.e., i.e. the image of $\bar{c}$ lies in the segment;

since geodesics of $\mathcal{Q}$ have constant speed, so does the horizontal lift, and $\bar{c}$ is the affine segment. Equality in the second inequality says $\bar{c}(1)$ attains the orbit distance, so $\bar{c}(1) = y^*$ by (a). Hence $\bar{c}$, and therefore $c = \pi \circ \bar{c}$, is uniquely determined and equals the geodesic of (b). Its initial velocity is $\bar{h}^*$, proving (5).

For the final statement: injectivity on $\{\|h\| < r_0\}$ holds by Lemma 2.6 (as $r_0 \leq \rho_0$), the map is a diffeomorphism onto its image, and the image contains $B([\theta_0], r_0)$ because every $[\theta']$ in that ball is $\text{Exp}_{[\theta_0]}(\bar{h}^*)$ with $\|\bar{h}^*\| = d_{\mathcal{Q}} < r_0$. $\square$

2.8 Bridge to Chapter 1's metric

Lemma 2.9 (Chapter 1 consistency is quotient-Riemannian consistency). *Let $\bar{d}$ be the Redner-style metric of Datta's Chapter 1, $\bar{d}([\theta'], [\theta_0]) = \min\{d(\theta', \theta_0), d(\theta', C) + d(\theta_0, C)\}$ with $C = \text{Orb}(\theta_0)$. Then*

$$\bar{d}([\theta'], [\theta_0]) \stackrel{(i)}{=} d(\theta', C) \stackrel{(ii)}{=} d(\theta_0, \text{Orb}(\theta')) \stackrel{(iii)}{=} d_{\mathcal{Q}}([\theta'], [\theta_0]) \quad \text{whenever the common value is } < r_0.$$

Consequently, Corollaries 1–2 of Datta's Chapter 1 yield $d_{\mathcal{Q}}([\hat{\theta}_n], [\theta_0]) \rightarrow 0$ almost surely (i.i.d. case) or in probability (weak-law case) with no additional argument.

Proof. (i): $\theta_0 \in C$ gives $d(\theta_0, C) = 0$ and $d(\theta', C) \leq d(\theta', \theta_0)$, so the minimum is $d(\theta', C)$. (ii): $d(\theta', C) = \min_R \|\theta' - \theta_0 \cdot R\| = \min_R \|\theta' \cdot R^{-1} - \theta_0\| = d(\theta_0, \text{Orb}(\theta'))$, using that the action is isometric. (iii): Lemma 2.3 at $\theta^* = \theta_0$ (its hypothesis $\delta < \rho_2(\theta_0)$ holds since $r_0 < \rho_2(\theta_0)$ by Lemma 2.7). The consistency transfer is immediate: Datta's results give $\bar{d} \rightarrow 0$, hence eventually $< r_0$, hence $d_{\mathcal{Q}} \rightarrow 0$. $\square$

2.9 Parallel transport

We use the standard fact that for a Riemannian submersion the Levi-Civita connection of the base is the horizontal projection of the total-space connection applied to horizontal lifts: for basic fields, $\nabla_X^{\mathcal{Q}} Y = P^{\mathcal{H}}(\nabla_X^{\Theta} \bar{Y})$ ([10, Lemma 1]; [2, Ch. 9]); the along-a-curve version follows by extending fields. Since Θ is flat, ∇^{Θ} is the ordinary directional derivative.

Proposition 2.10 (Transport ODE). *Let $\gamma(t) = \pi(\theta_0 + t\bar{\eta})$, $\bar{\eta} = (\eta_W, \eta_a) \in \mathcal{H}_{\theta_0}$, $\|\eta\| < \rho_1$, and write $W(t) = W_0 + t\eta_W$. A field $X(t)$ along γ with horizontal lifts $\bar{X}(t) = (A(t), a(t)) \in \mathcal{H}_{\gamma(t)}$ is parallel iff*

$$\dot{a}(t) = 0, \quad \dot{A}(t) = W(t)\Omega(t), \quad \mathbb{L}_{W(t)^{\top}W(t)}[\Omega(t)] = A(t)^{\top}\eta_W - \eta_W^{\top}A(t). \quad (6)$$

This linear ODE has a unique global solution on $[0, 1]$ for any initial value in $\mathcal{H}_{\theta_0}$; the solution stays horizontal, preserves the norm, and depends smoothly on $(t, \eta, \text{initial value})$.

Proof. By the fact quoted above, X is parallel ($\nabla_{\dot{\gamma}}^{\mathcal{Q}} X = 0$) iff the horizontal part of $\frac{d}{dt}\bar{X}$ vanishes, i.e. $\frac{d}{dt}\bar{X} \in \mathcal{V}_{\gamma(t)}$, i.e. $(\dot{A}, \dot{a}) = (W(t)\Omega(t), 0)$ for some $\Omega(t) \in \text{Skew}(q)$. Horizontality must persist; differentiating $\text{skew}(W(t)^{\top}A(t))$,

$$\frac{d}{dt} \text{skew}(W^{\top}A) = \text{skew}(\eta_W^{\top}A) + \text{skew}(W^{\top}W\Omega) = \frac{1}{2}(\eta_W^{\top}A - A^{\top}\eta_W) + \frac{1}{2}\mathbb{L}_{W^{\top}W}[\Omega],$$

using $2\text{skew}(W^{\top}W\Omega) = \mathbb{L}_{W^{\top}W}[\Omega]$ as in Lemma 2.1; this vanishes exactly under the Lyapunov equation in (6), whose right side is skew , so Lemma B.1 determines a unique $\Omega(t) \in \text{Skew}(q)$ depending linearly on $A(t)$ with $\|\Omega(t)\|_F \leq \|A(t)\|_F \|\eta_W\|_2 / (\ell_q - \rho_1)^2$. Hence (6) is a linear ODE with coefficients continuous in t and bounded on $[0, 1]$; Picard–Lindelöf gives a unique global solution, smooth

in all data. Conversely the solution has vertical derivative and, by the computation just made, remains horizontal, so it is the parallel transport. Norm preservation: $\frac{d}{dt}\|\bar{X}\|^2 = 2 \operatorname{tr}(\Omega^\top W^\top A) + 0 = 0$ because $W^\top A$ is symmetric and Ω skew. $\square$

2.10 The O'Neill tensor and a curvature bound

Proposition 2.11 (Curvature). *For horizontal $\xi = (A, a)$, $\eta = (B, b)$ at $\theta = (W, \sigma^2)$, the O'Neill tensor is*

$$A_\xi \eta = -\left(W \mathbb{L}_{W^\top W}^{-1}[A^\top B - B^\top A], 0\right), \quad (7)$$

and for orthonormal $\xi \perp \eta$,

$$0 \leq \sec_Q(d\pi \xi, d\pi \eta) = 3 \|A_\xi \eta\|^2 \leq \frac{3 \sigma_{\max}(W)^2}{\sigma_{\min}(W)^4};$$

in particular $\sec_Q \leq \kappa(r_0) := 3(\ell_1 + 2r_0)^2/(\ell_q - 2r_0)^4$ on $\bar{B}([\theta_0], r_0)$.

Proof. A is a tensor, so we may compute with any extensions. Extend ξ, η to the horizontal-projection fields $\bar{\xi}(\tilde{\theta}) := P_\theta^{\mathcal{H}}(A, a)$, $\bar{\eta}(\tilde{\theta}) := P_\theta^{\mathcal{H}}(B, b)$; these agree with ξ, η at θ (horizontality). In the flat total space, $A_\xi \eta$ is the vertical part of the directional derivative $D_\xi \bar{\eta}$ at θ ([10, p. 460]). Now $\bar{\eta}(\tilde{\theta}) = (B, b) - P_\theta^{\mathcal{V}}(B, b)$ and the constant part differentiates to zero, so $D_\xi \bar{\eta} = -D_\xi(P_\theta^{\mathcal{V}}(B, b))$. By Lemma 2.1, $P_\theta^{\mathcal{V}}(B, b) = (\widetilde{W} \Omega_B(\tilde{\theta}), 0)$ where Ω_B solves $\mathbb{L}_{\widetilde{W}^\top \widetilde{W}}[\Omega_B] = \widetilde{W}^\top B - B^\top \widetilde{W}$, and $\Omega_B(\theta) = 0$ because η is horizontal at θ . Differentiating along $\tilde{\theta} = \theta + s(A, a)$ at $s = 0$:

$$D_\xi(\widetilde{W} \Omega_B) = A \Omega_B(\theta) + W D\Omega_B[A] = W D\Omega_B[A],$$

and differentiating the Lyapunov identity, whose left side contributes $\mathbb{L}'[\Omega_B(\theta)] + \mathbb{L}_{W^\top W}[D\Omega_B[A]] = \mathbb{L}_{W^\top W}[D\Omega_B[A]]$ (the first term vanishes with $\Omega_B(\theta) = 0$) and whose right side differentiates to $A^\top B - B^\top A$:

$$\mathbb{L}_{W^\top W}[D\Omega_B[A]] = A^\top B - B^\top A.$$

Hence the vertical part of $D_\xi \bar{\eta}$ at θ is $-(W \mathbb{L}_{W^\top W}^{-1}[A^\top B - B^\top A], 0)$, which is (7). O'Neill's curvature formula for a flat total space ([10, Corollary 1]) gives, for orthonormal horizontal $\xi \perp \eta$, $\sec_Q = 3\|A_\xi \eta\|^2$. Finally, by Lemma B.1 and $\|A^\top B - B^\top A\|_F \leq 2\|A\|_F\|B\|_F \leq 2$,

$$\|A_\xi \eta\| \leq \sigma_{\max}(W) \cdot \frac{\|A^\top B - B^\top A\|_F}{2 \sigma_{\min}(W)^2} \leq \frac{\sigma_{\max}(W)}{\sigma_{\min}(W)^2},$$

and on the ball Weyl bounds the singular values as stated. $\square$

2.11 Consolidated caveats

Remark 2.12. (a) The action fixes the σ^2 -factor, so $\mathcal{Q}$ is the Riemannian product $(\mathbb{R}_*^{p \times q} / \mathrm{O}(q)) \times \mathbb{R}_{>0}$; the first factor, identified with rank- q positive-semidefinite matrices via $W \mapsto WW^\top$, is the quotient geometry studied by Massart and Absil [8], whose geodesics and Procrustes distance specialize the lemmas above. (b) $\mathcal{Q}$ is *incomplete* (boundaries as rank W drops or $\sigma^2 \rightarrow 0$); every $\operatorname{Exp} / \operatorname{Exp}^{-1} / \Pi$ statement in this paper is made on $\bar{B}([\theta_0], r_0)$, where Lemma 2.7 guarantees uniform normal neighborhoods, Proposition 2.11 bounds the curvature by $\kappa(r_0)$, and Proposition 2.8 makes the Procrustes rotation unique.

3 Asymptotic normality on $\mathcal{Q}$: definitions following Sun–Lin–Liu

We place ourselves verbatim in the framework of [13], with their parameter manifold taken to be $\mathcal{Q}$, their parameter of interest $\psi = \text{id}$ (so $\Psi = \mathcal{Q}$; their Assumption 3.1 on differentiability of ψ holds trivially with $\dot{\psi} = \text{id}$), and sample space $\mathcal{X} = \mathbb{R}^p$. Each definition reproduces theirs with only a change of symbols.

3.1 The statistical model on $\mathcal{Q}$

$f(x; \theta) = \mathcal{N}(x; 0, \Sigma(\theta))$ is $O(q)$ -invariant, so

$$\mathcal{P} = \{P_{[\theta]} : [\theta] \in \mathcal{Q}\}, \quad dP_{[\theta]}/dx = f(x; \theta),$$

is well defined and identifiable: $[\theta] \mapsto \Sigma(\theta)$ is injective (Datta’s Theorem 13 argument at an arbitrary base point; also a consequence of Theorem 4.2 plus connectedness, but the global statement is Datta’s). Write $r(x; [\theta]) = \sqrt{f(x; \theta)}$.

3.2 Well-posedness of the intrinsic estimation error

Let $\mathcal{C}([\theta]) \subset \mathcal{Q}$ collect the points not joined to $[\theta]$ by a unique minimizing geodesic contained in $\mathcal{Q}$ (for incomplete $\mathcal{Q}$ this generalized cut locus includes points with no minimizing geodesic); $\text{Exp}_{[\theta]}^{-1}$ is uniquely defined exactly off $\mathcal{C}([\theta])$.

Assumption 3.1 (cf. [13, Assumption 3.2]). $\sup_n \mathbb{P}_{\theta_0}([\hat{\theta}_n] \in \mathcal{C}([\theta_0])) = 0$.

Remark 3.2 (Operative error). Sun–Lin–Liu impose Assumption 3.1 so that the estimation error Exp^{-1} is almost surely defined. On our incomplete $\mathcal{Q}$ we proceed in two layers. First, define the *operative estimation error*

$$e_n := (\widehat{W}_n \widehat{R}_n - W_0, \widehat{\sigma}_n^2 - \sigma_0^2) \text{ when } \widehat{R}_n = \arg \min_{R \in O(q)} \|\widehat{W}_n R - W_0\|_F \text{ is unique, } e_n := 0 \text{ otherwise.}$$

By Proposition 2.8, $e_n = \text{Exp}_{[\theta_0]}^{-1}[\hat{\theta}_n]$ on the event $A_n := \{[\hat{\theta}_n] \in B([\theta_0], r_0)\}$, whose probability tends to one by Lemma 2.9; in particular $\mathbb{P}_{\theta_0}([\hat{\theta}_n] \in \mathcal{C}([\theta_0])) \rightarrow 0$, which suffices for all weak-convergence statements below. Second, Lemma 3.3 shows e_n is in fact almost surely given by the aligned difference for every fixed $n \geq p$, so the convention “ $e_n := 0$ ” is invoked only on a null set.

Lemma 3.3 (Almost-sure uniqueness of the alignment). *Under Assumption 5.1 with $n \geq p$, $\mathbb{P}_{\theta_0}(\widehat{R}_n \text{ is not unique}) = 0$.*

Proof. By Lemma B.3, $\widehat{R}_n$ is unique iff $W_0^\top \widehat{W}_n$ is nonsingular, a property invariant under $\widehat{W}_n \mapsto \widehat{W}_n R$. For $n \geq p$ the scaled Wishart matrix S_n has a Lebesgue density on $\text{Sym}(p)$; since the symmetric matrices with a repeated eigenvalue form a closed Lebesgue-null set (they have codimension two, von Neumann–Wigner), almost surely the eigenvalues $\delta_1 > \dots > \delta_p > 0$ of S_n are simple and positive. Then $\widehat{\sigma}_n^2 = (p - q)^{-1} \sum_{j>q} \delta_j \leq \delta_{q+1} < \delta_q$, so $(\Delta_q - \widehat{\sigma}_n^2 I_q)^{1/2}$ is invertible, and with $\widehat{W}_n = U_q(\Delta_q - \widehat{\sigma}_n^2 I_q)^{1/2} R$ (Tipping–Bishop),

$$W_0^\top \widehat{W}_n = V \Lambda U^\top U_q (\Delta_q - \widehat{\sigma}_n^2 I_q)^{1/2} R$$

is nonsingular iff $U^\top U_q$ is, iff $\varphi(S_n) := \det(U^\top P_q(S_n) U) \neq 0$, where $P_q(S) = U_q(S) U_q(S)^\top$ is the spectral projector onto the leading q -dimensional eigenspace and we used $\det(U^\top U_q)^2 = \det(U^\top P_q U)$.

Consider the open set $\mathcal{O} := \{S \in \text{Sym}(p) : \lambda_q(S) > \lambda_{q+1}(S)\}$. On $\mathcal{O}$, P_q is single-valued and real-analytic: with Γ a contour enclosing the top q eigenvalues and no others (locally constant in S), the Riesz formula $P_q(S) = (2\pi i)^{-1} \oint_{\Gamma} (zI - S)^{-1} dz$ exhibits P_q as an integral of the resolvent, analytic in S . Hence φ is real-analytic on $\mathcal{O}$.

$\mathcal{O}$ is connected: given $S = Q \text{diag}(\delta) Q^\top \in \mathcal{O}$ with $Q \in \text{O}(p)$ (flipping the sign of one column of Q if necessary to have $Q \in \text{SO}(p)$, which does not change S), first interpolate the ordered spectrum linearly to $\delta^* = (2, \dots, 2, 1, \dots, 1)$ (q twos); the entrywise interpolation of two ordered vectors is ordered and the gap at position q equals $(1-t)(\delta_q - \delta_{q+1}) + t > 0$, so the path stays in $\mathcal{O}$; then contract Q to I_p inside the connected group $\text{SO}(p)$, which fixes the spectrum. Finally, $\varphi \not\equiv 0$ on $\mathcal{O}$: at $S^\circ := U D^\circ U^\top + U_\perp U_\perp^\top$ with D° diagonal with distinct entries > 1 , the leading eigenspace is $\text{col}(U)$, so $\varphi(S^\circ) = \det(U^\top U U^\top U) = 1$.

The zero set of a real-analytic function, not identically zero, on a connected open set is Lebesgue-null [9]. Since S_n has a density and lies in $\mathcal{O}$ almost surely, $\mathbb{P}(\varphi(S_n) = 0) = 0$. $\square$

3.3 The definition of asymptotic normality

In [13] the estimation error of $\hat{\psi}_n$ is, per Table 1 and Definition 3.3 there,

$$(\text{linear space}) \quad \hat{\psi}_n - \psi(\theta) \quad \longrightarrow \quad (\text{manifold}) \quad \text{Exp}_{\psi(\theta)}^{-1} \hat{\psi}_n \in T_{\psi(\theta)} \Psi,$$

and their asymptotic-normality statements (Proposition 5.1, Corollary 5.1, Theorem 5.2, Appendix B.1) all take the form $\sqrt{n} \text{Exp}_{\theta_0}^{-1} \hat{\theta}_n \rightsquigarrow_P Z$, $Z \sim \mathcal{N}(0, V_{\theta_0})$ on the tangent space. We adopt exactly this notion.

Definition 3.4 (Asymptotic normality on $\mathcal{Q}$; cf. [13, Table 1, Prop. 5.1, Cor. 5.1]). Let $[\hat{\theta}_n]$ satisfy Assumption 3.1 (or the relaxation of Remark 3.2). The sequence $(\widehat{W}_n, \hat{\sigma}_n^2)$ is *asymptotically normal* at $[\theta_0]$ with asymptotic covariance operator $V_{[\theta_0]}$ if

$$\sqrt{n} \text{Exp}_{[\theta_0]}^{-1} [\hat{\theta}_n] \rightsquigarrow_{P_{\theta_0}} Z, \quad Z \in T_{[\theta_0]} \mathcal{Q}, \quad Z \sim \mathcal{N}(0, V_{[\theta_0]}), \quad (8)$$

with Z a tight Borel measurable random element of $T_{[\theta_0]} \mathcal{Q}$ (the phrasing of [13, Def. 4.3]) and $\mathcal{N}(0, V)$ the Gaussian measure on $T_{[\theta_0]} \mathcal{Q} \cong \mathcal{H}_{\theta_0}$ with covariance operator V .

Proposition 3.5 (Explicit Procrustes form). (8) holds iff $\sqrt{n} e_n \rightsquigarrow_{P_{\theta_0}} \mathcal{N}(0, V_{[\theta_0]})$ as random elements of $\mathcal{H}_{\theta_0} \subset \mathbb{R}^{p \times q} \times \mathbb{R}$.

Proof. On the events A_n of Remark 3.2, $e_n = \text{Exp}_{[\theta_0]}^{-1} [\hat{\theta}_n]$ and $\mathbb{P}(A_n) \rightarrow 1$. If $\zeta_n = \xi_n$ on A_n with $\mathbb{P}(A_n) \rightarrow 1$ and $\zeta_n \rightsquigarrow Z$, then for every bounded Lipschitz φ , $|\mathbb{E}\varphi(\xi_n) - \mathbb{E}\varphi(\zeta_n)| \leq 2\|\varphi\|_\infty \mathbb{P}(A_n^c) \rightarrow 0$, so $\xi_n \rightsquigarrow Z$ as well; apply this in both directions. $\square$

Remark 3.6 (Geometric content; chart invariance). Proposition 3.5 is the well-posed replacement of the false naive CLT: *after* optimal rotational alignment the ordinary error is asymptotically Gaussian, supported on the $(pq + 1 - \frac{q(q-1)}{2})$ -dimensional horizontal space; the $\frac{q(q-1)}{2}$ vertical directions carry no mass. As a measure on $\mathbb{R}^{pq+1}$ the limit is degenerate — by necessity, not by defect. The definition is chart-independent: if φ is any smooth chart centered at $[\theta_0]$ with differential L on $T_{[\theta_0]} \mathcal{Q}$, then, since $\text{Exp}_{[\theta_0]}^{-1}$ is itself a chart (Proposition 2.8) with differential the identity, the delta method shows (8) is equivalent to $\sqrt{n}(\varphi([\hat{\theta}_n]) - \varphi([\theta_0])) \rightsquigarrow \mathcal{N}(0, LVL^*)$; the covariance transforms as a $(2,0)$ -tensor. Section 7 develops the geometric and statistical reading at length.

3.4 Companion definitions from [13] ($\Psi = \mathcal{Q}$, $\psi = \text{id}$)

Definition 3.7 (DQM and score operator; cf. [13, Def. 3.1]). For $h \in T_{[\theta]} \mathcal{Q}$ set $[\theta]_h := \text{Exp}_{[\theta]}(h)$. The model $\mathcal{P}$ is *differentiable in quadratic mean at $P_{[\theta]}$* if there exists $s(\cdot; [\theta]) : \mathcal{X} \rightarrow T_{[\theta]}^* \mathcal{Q}$ with, as $h \rightarrow 0$,

$$\int_{\mathcal{X}} \left(r(x; [\theta]_h) - r(x; [\theta]) - \frac{1}{2} s(x; [\theta])(h) r(x; [\theta]) \right)^2 dx = o(\|h\|^2). \quad (9)$$

Write $S_{[\theta]} := s(X; [\theta])$; the model tangent space is $\Lambda = \text{cl}[\text{span}\{S_{[\theta]}(h)\}]$.

Definition 3.8 (Fisher information operator; cf. [13, Def. 3.2]). $G_{[\theta]}(\nu_1, \nu_2) := \mathbb{E}[S_{[\theta]}(\nu_1)S_{[\theta]}(\nu_2)]$, i.e. $G_{[\theta]} = \mathbb{E}[S_{[\theta]} \otimes S_{[\theta]}]$, with inverse $G_{[\theta]}^{-1}$ when nonsingular.

Definition 3.9 (Locally regular estimators; cf. [13, Def. 3.5]). For $h \in T_{[\theta_0]} \mathcal{Q}$ set $[\theta]_{n,h} := \text{Exp}_{[\theta_0]}(h/\sqrt{n})$. The sequence $[\hat{\theta}_n]$ is *(locally) regular at $[\theta_0]$* if there is $Z_{[\theta_0]}$ with law independent of h such that

$$\Pi_{[\theta]_{n,h}}^{[\theta_0]} \sqrt{n} \text{Exp}_{[\theta]_{n,h}}^{-1} [\hat{\theta}_n] \rightsquigarrow_{P_{[\theta]_{n,h}}} Z_{[\theta_0]},$$

with Π the parallel transport along the minimizing geodesic, exactly as in [13] (computable here by Proposition 2.10).

Definition 3.10 (Asymptotic linearity; cf. [13, Thm. 3.2]). $[\hat{\theta}_n]$ is *asymptotically linear at $[\theta_0]$* with influence operator $\text{IF} \in T_{[\theta_0]} \mathcal{Q}$ (square-integrable, mean zero) if $\sqrt{n} \text{Exp}_{[\theta_0]}^{-1} [\hat{\theta}_n] = n^{-1/2} \sum_{i=1}^n \text{IF}(x_i) + o_{P_{\theta_0}}(1)$. If in addition Definition 3.9 holds, the calculus of influence operators [13, Thm. 3.2] forces $\text{id} = \dot{\psi} = \mathbb{E}[\text{IF} \cdot S_{[\theta_0]}]$.

An asymptotically normal, regular sequence is *efficient* if its limit is the pure Gaussian factor $\mathcal{N}(0, G^{-1})$ in the convolution theorem [13, Thm. 3.1], equivalently attains the LAM bound [13, Thm. 3.3].

3.5 The vocabulary of [13, Table 1] at PPCA

concept	Sun–Lin–Liu (manifold)	this draft ($\mathcal{Q} = \Theta / \mathcal{O}(q)$)
perturbation / path	$\theta_h = \text{Exp}_{\theta} h$	$[\theta_0]_h = [(W_0, \sigma_0^2) + \bar{h}]$ (Proposition 2.5)
estimation error	$\text{Exp}_{\psi(\theta)}^{-1} \hat{\psi}_n$	$e_n = (\widehat{W}_n \widehat{R}_n - W_0, \widehat{\sigma}_n^2 - \sigma_0^2)$ (Proposition 2.8, Lemma 3.3)
asymptotic normality	$\sqrt{n} \text{Exp}_{\theta_0}^{-1} \hat{\theta}_n \rightsquigarrow \mathcal{N}(0, V)$	$\sqrt{n} e_n \rightsquigarrow \mathcal{N}(0, G^{-1})$ on $\mathcal{H}_{\theta_0}$ (Theorem 5.3)
regular estimator	$\Pi_{\psi(\theta_{n,h})}^{\psi(\theta)} \sqrt{n} \text{Exp}_{\psi(\theta_{n,h})}^{-1} \hat{\psi}_n \rightsquigarrow Z_{\theta}$	same, Π from Proposition 2.10
score / Fisher	$s : \mathcal{X} \rightarrow T_{\theta}^* \Theta$, $G = \mathbb{E}[S \otimes S]$	(10)–(11); explicit blocks in Propositions 4.4, 4.5
influence relation	$\dot{\psi} = \mathbb{E}[\text{IF} \cdot S]$	$\text{IF} = G^{-1}s$; components in Corollary 4.7 and (15)

4 The Fisher geometry of the PPCA model

Throughout, $\dot{\Sigma}[A, a] := AW_0^\top + W_0A^\top + aI_p$ denotes the differential of $\theta \mapsto \Sigma(\theta)$ at θ_0 (immediate from $\Sigma(\theta_0 + t(A, a)) = \Sigma_0 + t\dot{\Sigma}[A, a] + t^2AA^\top$, an *exact* quadratic expansion used again in Step 5).

4.1 Score and information

Lemma 4.1 (Score and Fisher operator). *For $h \in T_{[\theta_0]}\mathcal{Q}$ with lift $\bar{h}$, define*

$$s(x)(h) := \frac{1}{2} \operatorname{tr} \left(\Sigma_0^{-1} (xx^\top - \Sigma_0) \Sigma_0^{-1} \dot{\Sigma}[\bar{h}] \right), \quad (10)$$

$$G_{[\theta_0]}(h, h') := \frac{1}{2} \operatorname{tr} \left(\Sigma_0^{-1} \dot{\Sigma}[\bar{h}] \Sigma_0^{-1} \dot{\Sigma}[\bar{h}'] \right). \quad (11)$$

Then $h \mapsto s(x)(h)$ is linear (an element of $T_{[\theta_0]}^\mathcal{Q}$), $\mathbb{E}s(X)(h) = 0$, and $\mathbb{E}[s(X)(h)s(X)(h')] = G_{[\theta_0]}(h, h')$.*

Proof. Linearity is clear since $\bar{h} \mapsto \dot{\Sigma}[\bar{h}]$ is linear. Mean zero: $\mathbb{E}xx^\top = \Sigma_0$. For the covariance, write $s(h) = \frac{1}{2}x^\top M_h x - \frac{1}{2} \operatorname{tr}(\Sigma_0 M_h)$ with $M_h := \Sigma_0^{-1} \dot{\Sigma}[\bar{h}] \Sigma_0^{-1} \in \operatorname{Sym}(p)$; by Lemma A.1,

$$\mathbb{E}[s(h)s(h')] = \frac{1}{4} \operatorname{Cov}(x^\top M_h x, x^\top M_{h'} x) = \frac{1}{4} \cdot 2 \operatorname{tr}(M_h \Sigma_0 M_{h'} \Sigma_0) = \frac{1}{2} \operatorname{tr}(\Sigma_0^{-1} \dot{\Sigma}[\bar{h}] \Sigma_0^{-1} \dot{\Sigma}[\bar{h}']). \quad \square$$

4.2 Nondegeneracy without eigengap

Theorem 4.2 (The crux). *Under $1 \leq q < p$, $\operatorname{rank}(W_0) = q$, $\sigma_0^2 > 0$: $\ker \dot{\Sigma} = \mathcal{V}_{\theta_0}$ exactly; consequently $G_{[\theta_0]}$ is positive definite on $T_{[\theta_0]}\mathcal{Q} \cong \mathcal{H}_{\theta_0}$, and $\iota : [\theta] \mapsto \Sigma(\theta)$ is an injective immersion of $\mathcal{Q}$ into $\operatorname{Sym}(p)$. No condition on the multiplicities of $\ell_1, \dots, \ell_q$ is used.*

Proof. Every $A \in \mathbb{R}^{p \times q}$ decomposes uniquely as $A = UBV^\top + U_\perp CV^\top$ with $B = U^\top AV \in \mathbb{R}^{q \times q}$, $C = U_\perp^\top AV \in \mathbb{R}^{(p-q) \times q}$. Then $AW_0^\top = (UBV^\top + U_\perp CV^\top)(V\Lambda U^\top) = UBAU^\top + U_\perp CAU^\top$ and $W_0A^\top = (AW_0^\top)^\top = U\Lambda B^\top U^\top + U\Lambda C^\top U_\perp^\top$, so in the $(U, U_\perp)$ block basis

$$\dot{\Sigma}[A, a] = \begin{pmatrix} B\Lambda + \Lambda B^\top + aI_q & \Lambda C^\top \\ C\Lambda & aI_{p-q} \end{pmatrix}. \quad (12)$$

Suppose $\dot{\Sigma}[A, a] = 0$. The $(U_\perp, U_\perp)$ block gives $aI_{p-q} = 0$, so $a = 0$ (here $q < p$ enters). The $(U_\perp, U)$ block gives $C\Lambda = 0$, so $C = 0$ (Λ invertible). The (U, U) block gives $B\Lambda + \Lambda B^\top = 0$; multiplying on the left and on the right by Λ^{-1} ,

$$\Lambda^{-1}B + B^\top \Lambda^{-1} = 0 \iff \tilde{\Omega} := \Lambda^{-1}B \in \operatorname{Skew}(q).$$

Then $A = UBV^\top = U\Lambda\tilde{\Omega}V^\top = U\Lambda V^\top(V\tilde{\Omega}V^\top) = W_0\Omega$, $\Omega := V\tilde{\Omega}V^\top \in \operatorname{Skew}(q)$; i.e. $(A, a) \in \mathcal{V}_{\theta_0}$, cf. (1). Conversely $\dot{\Sigma}[W_0\Omega, 0] = W_0\Omega W_0^\top + W_0\Omega^\top W_0^\top = W_0(\Omega + \Omega^\top)W_0^\top = 0$, so $\ker \dot{\Sigma} = \mathcal{V}_{\theta_0}$; the dimensions match: $\dim \ker = \frac{q(q-1)}{2} = \dim \mathcal{V}_{\theta_0}$. Finally, for $\bar{h} \in \mathcal{H}_{\theta_0}$,

$$G(h, h) = \frac{1}{2} \|\Sigma_0^{-1/2} \dot{\Sigma}[\bar{h}] \Sigma_0^{-1/2}\|_F^2 = 0 \iff \dot{\Sigma}[\bar{h}] = 0 \iff \bar{h} \in \mathcal{V}_{\theta_0} \cap \mathcal{H}_{\theta_0} = \{0\},$$

and injectivity of $d\iota = \dot{\Sigma}|_{\mathcal{H}}$ at every point of $\mathcal{Q}$ (only $\operatorname{rank} W = q$, $q < p$ were used) makes ι an immersion; it is injective by identifiability. $\square$

4.3 Block coordinates on the horizontal space

Lemma 4.3 (Linear coordinates). *The map*

$$\Phi : \mathcal{H}_{\theta_0} \rightarrow \text{Sym}(q) \times \mathbb{R}^{(p-q) \times q} \times \mathbb{R}, \quad (A, a) \mapsto (S, C, a), \quad S := \text{sym}(B\Lambda), \quad B = U^\top AV, \quad C = U_\perp^\top AV,$$

is a linear isomorphism.

Proof. Horizontality (2) reads $W_0^\top A = V\Lambda BV^\top \in \text{Sym}(p) \iff \Lambda B \in \text{Sym}(q)$. On the subspace $\{B : \Lambda B \in \text{Sym}(q)\}$ — of dimension $q(q+1)/2$, since $B \mapsto \Lambda B$ is a linear isomorphism of $\mathbb{R}^{q \times q}$ carrying it onto $\text{Sym}(q)$ — consider $B \mapsto \text{sym}(B\Lambda)$. If $\text{sym}(B\Lambda) = 0$ and $\Lambda B \in \text{Sym}(q)$, then $B^\top = \Lambda B\Lambda^{-1}$ and $B\Lambda = -\Lambda B^\top$; substituting the first into the second, $B\Lambda = -\Lambda^2 B\Lambda^{-1}$, i.e. $B\Lambda^2 + \Lambda^2 B = 0$, whence $B = 0$ by Lemma B.1 ($\Lambda^2 \succ 0$). Thus $B \mapsto \text{sym}(B\Lambda)$ is injective between spaces of equal dimension $q(q+1)/2$, hence bijective; C and a are unconstrained, and $\dim \mathcal{H}_{\theta_0} = \frac{q(q+1)}{2} + (p-q)q + 1 = pq + 1 - \frac{q(q-1)}{2}$, matching (2). $\square$

4.4 The explicit Fisher operator

Proposition 4.4 (Explicit G). *In the coordinates of Lemma 4.3, with $M_{11} := 2S + aI_q$ (note $B\Lambda + \Lambda B^\top = 2\text{sym}(B\Lambda) = 2S$),*

$$G(h, h) = \underbrace{\frac{1}{2} \text{tr}((D^{-1}M_{11})^2)}_{(S,a)\text{-block}} + \underbrace{\sum_{i=1}^q \frac{\ell_i^2}{\sigma_0^2 \lambda_i} \|Ce_i\|^2}_{C\text{-block}} + \underbrace{\frac{(p-q)a^2}{2\sigma_0^4}}_{\text{noise block}}, \quad (13)$$

with no cross terms between the C -block and the (S, a) -block. In scalar coordinates,

$$G(h, h) = \sum_{i=1}^q \frac{(2S_{ii} + a)^2}{2\lambda_i^2} + \sum_{i < j} \frac{4S_{ij}^2}{\lambda_i \lambda_j} + \sum_{i=1}^q \frac{\ell_i^2}{\sigma_0^2 \lambda_i} \|Ce_i\|^2 + \frac{(p-q)a^2}{2\sigma_0^4}.$$

Proof. With (12) and $\Sigma_0^{-1} = UD^{-1}U^\top + \sigma_0^{-2}U_\perp U_\perp^\top$,

$$\Sigma_0^{-1} \dot{\Sigma}[\bar{h}] = \begin{pmatrix} D^{-1}M_{11} & D^{-1}\Lambda C^\top \\ \sigma_0^{-2}C\Lambda & \sigma_0^{-2}aI_{p-q} \end{pmatrix},$$

whose square has diagonal blocks $D^{-1}M_{11}D^{-1}M_{11} + \sigma_0^{-2}D^{-1}\Lambda C^\top C\Lambda$ and $\sigma_0^{-2}C\Lambda D^{-1}\Lambda C^\top + \sigma_0^{-4}a^2I_{p-q}$. Taking traces and using cyclicity together with $\Lambda D^{-1} = D^{-1}\Lambda$ (diagonal),

$$\text{tr}((\Sigma_0^{-1} \dot{\Sigma})^2) = \text{tr}((D^{-1}M_{11})^2) + 2\sigma_0^{-2} \text{tr}(\Lambda D^{-1}\Lambda C^\top C) + (p-q)a^2\sigma_0^{-4}.$$

Halving gives (13). For the scalar form: $\text{tr}((D^{-1}M)^2) = \sum_{i,j} M_{ij}M_{ji}/(\lambda_i\lambda_j) = \sum_{i,j} M_{ij}^2/(\lambda_i\lambda_j)$ for symmetric $M = 2S + aI_q$, which splits into the diagonal terms $(2S_{ii} + a)^2/\lambda_i^2$ and the off-diagonal terms $2\sum_{i < j} 4S_{ij}^2/(\lambda_i\lambda_j)$, halved as stated; and $(\Lambda D^{-1}\Lambda)_{ii} = \ell_i^2/\lambda_i$ gives the C -term. There are no (S, a) - C cross terms because those arise only from products of a diagonal with an off-diagonal block, which are off-diagonal and traceless in this block pattern. $\square$

4.5 The explicit inverse: limiting covariances

Proposition 4.5 (Explicit G^{-1}). *Write $Z = (Z_S, Z_C, Z_a) \sim \mathcal{N}(0, G^{-1})$ in the coordinates of Lemma 4.3. Then the C -block is independent of everything else, its entries are mutually independent, the off-diagonal S -entries are mutually independent and independent of $(\text{diag } Z_S, Z_a)$, and*

$$\begin{aligned} \text{Var}(Z_{C_{ki}}) &= \frac{\sigma_0^2 \lambda_i}{\ell_i^2} \quad (1 \leq k \leq p-q), & \text{Var}(Z_{S_{ij}}) &= \frac{\lambda_i \lambda_j}{4} \quad (i < j), \\ \text{Var}(Z_a) &= \frac{2\sigma_0^4}{p-q}, & \text{Cov}(Z_a, Z_{S_{ii}}) &= -\frac{\sigma_0^4}{p-q}, \\ \text{Cov}(Z_{S_{ii}}, Z_{S_{jj}}) &= \delta_{ij} \frac{\lambda_i^2}{2} + \frac{\sigma_0^4}{2(p-q)}. \end{aligned}$$

Proof. Independence across blocks and within the decoupled coordinates: (13) is a sum of quadratic forms in disjoint variables, so G is block diagonal in the coordinates $\{S_{ij}\}_{i < j} \cup \{C_{ki}\} \cup \{S_{11}, \dots, S_{qq}, a\}$, with scalar diagonal entries $4/(\lambda_i \lambda_j)$ and $\ell_i^2/(\sigma_0^2 \lambda_i)$ on the first two groups; their inverses are the stated variances. For the coupled block $x = (S_{11}, \dots, S_{qq}, a)$, expanding $\sum_i (2S_{ii} + a)^2 / (2\lambda_i^2) + (p-q)a^2 / (2\sigma_0^4)$ gives the Fisher matrix

$$\mathfrak{G} = \begin{pmatrix} \text{diag}(2/\lambda_i^2) & v \\ v^\top & g \end{pmatrix}, \quad v_i = \frac{1}{\lambda_i^2}, \quad g = \frac{1}{2} \sum_i \frac{1}{\lambda_i^2} + \frac{p-q}{2\sigma_0^4}.$$

Block inversion: the Schur complement of the S -block is

$$s = g - v^\top \text{diag}(\lambda_i^2/2) v = g - \frac{1}{2} \sum_i \frac{1}{\lambda_i^2} = \frac{p-q}{2\sigma_0^4},$$

so $\text{Var}(Z_a) = s^{-1} = 2\sigma_0^4/(p-q)$; $\text{Cov}(Z_a, Z_{S_{ii}}) = -(\text{diag}(\lambda_i^2/2)v)_i s^{-1} = -\frac{1}{2} \cdot \frac{2\sigma_0^4}{p-q} = -\frac{\sigma_0^4}{p-q}$; and $\text{Cov}(Z_{S_{ii}}, Z_{S_{jj}}) = \delta_{ij} \frac{\lambda_i^2}{2} + \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{2\sigma_0^4}{p-q} = \delta_{ij} \frac{\lambda_i^2}{2} + \frac{\sigma_0^4}{2(p-q)}$, by the standard formulas $(\mathfrak{G}^{-1})_{SS} = \text{diag}(\lambda_i^2/2) + \text{diag}(\lambda_i^2/2) v s^{-1} v^\top \text{diag}(\lambda_i^2/2)$ and $(\mathfrak{G}^{-1})_{Sa} = -\text{diag}(\lambda_i^2/2) v s^{-1}$, with $\text{diag}(\lambda_i^2/2) v = (1/2, \dots, 1/2)^\top$. $\square$

Remark 4.6 (Moment verification of (13)). The score coordinates computed in (14) below satisfy, with $y = U^\top x \sim \mathcal{N}(0, D)$ and $z = U_\perp^\top x \sim \mathcal{N}(0, \sigma_0^2 I)$ independent: $\mathbb{E}[s_{S_{ii}}^2] = \text{Var}(y_i^2)/\lambda_i^4 = 2/\lambda_i^2$; $\mathbb{E}[s_{S_{ij}}^2] = 4\mathbb{E}[y_i^2 y_j^2]/(\lambda_i \lambda_j)^2 = 4/(\lambda_i \lambda_j)$; $\mathbb{E}[s_{C_{ki}}^2] = \sigma_0^{-4} \ell_i^2 \lambda_i^{-2} \mathbb{E}[y_i^2 z_k^2] = \ell_i^2/(\sigma_0^2 \lambda_i)$; $\mathbb{E}[s_{S_{ii}} s_a] = \frac{1}{2} \mathbb{E}[s_{S_{ii}}^2] = 1/\lambda_i^2$; $\mathbb{E}[s_a^2] = \frac{1}{4} [\sum_j 2/\lambda_j^2 + 2(p-q)/\sigma_0^4] = \frac{1}{2} \sum_j \lambda_j^{-2} + (p-q)/(2\sigma_0^4)$; and all products involving exactly one z -odd factor vanish. These are exactly the entries of $\mathfrak{G}$ in Propositions 4.4 and 4.5 — an independent confirmation of the block computation.

4.6 The noise variance: closed form, influence function, and an independent derivation

We first record the score in coordinates. With $y = U^\top x$, $z = U_\perp^\top x$, the matrix $E := \Sigma_0^{-1}(xx^\top - \Sigma_0)\Sigma_0^{-1}$ has blocks

$$E_{11} = D^{-1}(yy^\top - D)D^{-1}, \quad E_{12} = \sigma_0^{-2} D^{-1} y z^\top, \quad E_{21} = E_{12}^\top, \quad E_{22} = \sigma_0^{-4} (zz^\top - \sigma_0^2 I_{p-q}),$$

so that, pairing with (12), $2s(x)(h) = \text{tr}(E\dot{\Sigma}[\bar{h}]) = \text{tr}(E_{11}M_{11}) + 2\sigma_0^{-2}z^\top C(\Lambda D^{-1})y + a \text{tr}(E_{22})$, and therefore

$$s_{S_{ii}} = \frac{y_i^2 - \lambda_i}{\lambda_i^2}, \quad s_{S_{ij}} = \frac{2y_i y_j}{\lambda_i \lambda_j} \quad (i < j), \quad s_{C_{ki}} = \frac{\ell_i y_i z_k}{\sigma_0^2 \lambda_i}, \quad s_a = \frac{1}{2} \left[\sum_i \frac{y_i^2 - \lambda_i}{\lambda_i^2} + \frac{\|z\|^2 - (p-q)\sigma_0^2}{\sigma_0^4} \right]. \quad (14)$$

Corollary 4.7 (Noise variance). *Under Theorem 5.3,*

$$\sqrt{n}(\hat{\sigma}_n^2 - \sigma_0^2) \rightsquigarrow \mathcal{N}\left(0, \frac{2\sigma_0^4}{p-q}\right), \quad \text{IF}_{\sigma^2}(x) = \frac{\|U_\perp^\top x\|^2 - (p-q)\sigma_0^2}{p-q}.$$

Proof. The σ^2 -component of $\text{Exp}_{[\theta_0]}^{-1}[\hat{\theta}_n]$ is *exactly* the coordinate a (Proposition 2.8), so the asymptotic variance is $\text{Var}(Z_a)$ of Proposition 4.5. For the influence function, $\text{IF} = G^{-1}s$ componentwise gives, using Proposition 4.5 and (14),

$$\begin{aligned} \text{IF}_a &= \text{Var}(Z_a) s_a + \sum_i \text{Cov}(Z_a, Z_{S_{ii}}) s_{S_{ii}} \\ &= \frac{2\sigma_0^4}{p-q} \cdot \frac{1}{2} \left[\sum_i \frac{y_i^2 - \lambda_i}{\lambda_i^2} + \frac{\|z\|^2 - (p-q)\sigma_0^2}{\sigma_0^4} \right] - \frac{\sigma_0^4}{p-q} \sum_i \frac{y_i^2 - \lambda_i}{\lambda_i^2} = \frac{\|z\|^2 - (p-q)\sigma_0^2}{p-q}, \end{aligned}$$

the y -terms cancelling identically. Consistency check: $\text{Var}(\text{IF}_{\sigma^2}) = \text{Var}(\|z\|^2)/(p-q)^2 = 2(p-q)\sigma_0^4/(p-q)^2 = 2\sigma_0^4/(p-q)$, matching $\text{Var}(Z_a)$, and $\mathbb{E}[\text{IF}_{\sigma^2} s_a] = \frac{1}{2} \cdot 2(p-q)\sigma_0^4/((p-q)\sigma_0^4) = 1$, as the influence relation $\mathbb{E}[\text{IF} \cdot S] = \text{id}$ requires on the a -coordinate. $\square$

Lemma 4.8 (Independent derivation via eigenvalue perturbation). *Let $P := U_\perp U_\perp^\top$ and $\hat{P}$ the spectral projector onto the trailing $(p-q)$ -dimensional eigenspace of S_n . Then*

$$0 \leq \text{tr}(PS_n) - \sum_{j>q} \delta_j \leq \|P - \hat{P}\|_F \|S_n - \Sigma_0\|_F \leq \frac{2\sqrt{2}}{\ell_q^2} \|S_n - \Sigma_0\|_F^2 = O_P(n^{-1}),$$

and consequently $\sqrt{n}(\hat{\sigma}_n^2 - \sigma_0^2) = n^{-1/2} \sum_{i=1}^n \text{IF}_{\sigma^2}(x_i) + o_P(1)$ with IF_{σ^2} as in Corollary 4.7 — an independent confirmation not using G^{-1} .

Proof. By Ky Fan's variational principle, $\sum_{j>q} \delta_j = \min_Q \text{tr}(QS_n)$ over rank- $(p-q)$ orthoprojectors, attained at $\hat{P}$; likewise $(p-q)\sigma_0^2 = \min_Q \text{tr}(Q\Sigma_0)$ is attained at P (its minimizer, unique because of the population gap $\ell_q^2 > 0$). Hence $\text{tr}(PS_n) - \text{tr}(\hat{P}S_n) \geq 0$, and

$$\text{tr}(PS_n) - \text{tr}(\hat{P}S_n) = \text{tr}((P - \hat{P})(S_n - \Sigma_0)) + \underbrace{\text{tr}(P\Sigma_0) - \text{tr}(\hat{P}\Sigma_0)}_{\leq 0} \leq \|P - \hat{P}\|_F \|S_n - \Sigma_0\|_F$$

by Cauchy–Schwarz. The Davis–Kahan inequality in the form of [16, Thm. 2], with population gap $\lambda_q(\Sigma_0) - \lambda_{q+1}(\Sigma_0) = \ell_q^2$, gives $\|\hat{P} - P\|_F \leq 2\sqrt{2} \|S_n - \Sigma_0\|_F / \ell_q^2$. Since each entry of $S_n - \Sigma_0$ has variance $O(1/n)$, $\mathbb{E}\|S_n - \Sigma_0\|_F^2 = O(1/n)$ and the product is $O_P(1/n)$. Therefore

$$\sqrt{n} \left(\hat{\sigma}_n^2 - \frac{\text{tr}(PS_n)}{p-q} \right) = \frac{\sqrt{n}}{p-q} \left(\sum_{j>q} \delta_j - \text{tr}(PS_n) \right) = O_P(n^{-1/2}) \rightarrow 0,$$

while $\text{tr}(PS_n)/(p-q) - \sigma_0^2 = n^{-1} \sum_i (\|U_\perp^\top x_i\|^2 - (p-q)\sigma_0^2)/(p-q)$ *exactly*. $\square$

Similarly, the C -block of $\text{IF} = G^{-1}s$ is, from Proposition 4.5 and (14),

$$\text{IF}_{C_{ki}}(x) = \frac{\sigma_0^2 \lambda_i}{\ell_i^2} \cdot \frac{\ell_i y_i z_k}{\sigma_0^2 \lambda_i} = \frac{z_k y_i}{\ell_i}, \quad \mathbb{E} \text{IF}_{C_{ki}}^2 = \frac{\mathbb{E}[y_i^2] \mathbb{E}[z_k^2]}{\ell_i^2} = \frac{\sigma_0^2 \lambda_i}{\ell_i^2} \checkmark. \quad (15)$$

Remark 4.9 (Three cross-checks against classical PCA asymptotics). (1) *Pooled noise eigenvalues.* $\text{Var}(Z_a) = 2\sigma_0^4/(p-q)$ matches the classical variance of the average of $p-q$ sample eigenvalues attached to a population eigenvalue of multiplicity $p-q$ (Lemma 4.8 is that classical computation). (2) *Leading-block entries.* By (12) the error of $(U^\top \widehat{\Sigma}_n U)_{ii}$ is $2Z_{S_{ii}} + Z_a$, with

$$\text{Var}(2Z_{S_{ii}} + Z_a) = 4\left(\frac{\lambda_i^2}{2} + \frac{\sigma_0^4}{2(p-q)}\right) + \frac{2\sigma_0^4}{p-q} + 4\left(-\frac{\sigma_0^4}{p-q}\right) = 2\lambda_i^2 = \text{Var}((U^\top x)_i^2),$$

exactly the sample-covariance variance for this coordinate — the constrained estimator does not beat S_n there, as it must not. (3) *Anderson's eigenvector terms.* When the ℓ_i are simple, the i -th aligned loading column is $\ell_i u_i + U B e_i + U_\perp C e_i$, so the noise-subspace tilt of the estimated direction u_i is $U_\perp Z_C e_i / \ell_i$, with per-coordinate variance $\sigma_0^2 \lambda_i / \ell_i^4 = \lambda_i \lambda_j / (\lambda_i - \lambda_j)^2$ for $\lambda_j = \sigma_0^2$ — Anderson's formula [1] for the noise directions. With repeated ℓ_i 's the eigenvector-wise reading fails, but Proposition 4.5 is untouched: the quotient-level limit does not see the multiplicities.

5 Main theorems

Assumption 5.1. $1 \leq q < p$, $\text{rank}(W_0) = q$, $\sigma_0^2 > 0$; $x_1, \dots, x_n$ i.i.d. from $P_{[\theta_0]}$.

Theorem 5.2 (DQM and LAN of the quotient model). *Under Assumption 5.1, the model $\mathcal{P}$ satisfies Definition 3.7 at every $[\theta] \in \mathcal{Q}$, with score operator (10) and Fisher operator (11); consequently, for $[\theta]_{n,h} = \text{Exp}_{[\theta_0]}(h/\sqrt{n})$,*

$$\sum_{i=1}^n \log \frac{f(x_i; [\theta]_{n,h})}{f(x_i; [\theta_0])} = \frac{1}{\sqrt{n}} \sum_{i=1}^n s(x_i)(h) - \frac{1}{2} G(h, h) + o_{P_{\theta_0}}(1), \quad (16)$$

and the product experiments $\{P_{[\theta]_{n,h}}^{\otimes n}\}$ and $\{P_{[\theta_0]}^{\otimes n}\}$ are mutually contiguous.

Theorem 5.3 (Asymptotic linearity and CLT for the MLE). *Under Assumption 5.1, the Tipping-Bishop MLE $\widehat{\theta}_n = (\widehat{W}_n, \widehat{\sigma}_n^2)$ satisfies Definition 3.10 with $\text{IF} = G_{[\theta_0]}^{-1}s$:*

$$\sqrt{n} \text{Exp}_{[\theta_0]}^{-1}[\widehat{\theta}_n] = \frac{1}{\sqrt{n}} \sum_{i=1}^n G_{[\theta_0]}^{-1}s(x_i) + o_{P_{\theta_0}}(1) \rightsquigarrow_{P_{\theta_0}} \mathcal{N}(0, G_{[\theta_0]}^{-1}) \quad \text{on } T_{[\theta_0]}\mathcal{Q},$$

with the block description of $\mathcal{N}(0, G^{-1})$ in Proposition 4.5. Equivalently (Proposition 3.5), $\sqrt{n}(\widehat{W}_n \widehat{R}_n - W_0, \widehat{\sigma}_n^2 - \sigma_0^2) \rightsquigarrow \mathcal{N}(0, G^{-1})$ supported on $\mathcal{H}_{\theta_0}$.

Theorem 5.4 (Regularity and efficiency). *Under Assumption 5.1: (i) the MLE is locally regular at $[\theta_0]$ (Definition 3.9); (ii) every regular estimator sequence has limit law $\mathcal{N}(0, G^{-1}) * \Delta$ with Δ independent of the Gaussian factor ([13, Thm. 3.1] with $\psi = \text{id}$); (iii) the LAM bound of [13, Thm. 3.3] holds with limiting covariance G^{-1} . Hence the MLE is asymptotically efficient on $\mathcal{Q}$.*

Corollary 5.5 (Covariance-level CLT). $\widehat{\Sigma}_n := \widehat{W}_n \widehat{W}_n^\top + \widehat{\sigma}_n^2 I_p$ satisfies

$$\sqrt{n}(\widehat{\Sigma}_n - \Sigma_0) \rightsquigarrow \mathcal{N}(0, i G^{-1} i^*) \quad \text{in } \text{Sym}(p), \quad i = \dot{\Sigma}|_{\mathcal{H}_{\theta_0}},$$

and $\widehat{\Sigma}_n = U_q \Delta_q U_q^\top + \widehat{\sigma}_n^2 (I_p - U_q U_q^\top)$: the constrained MLE keeps the top q eigenpairs of S_n and pools the rest.

Corollary 5.6 (Blockwise comparison with the sample covariance). *In the $(U, U_\perp)$ basis, the limit of $\sqrt{n}(\widehat{\Sigma}_n - \Sigma_0)$ is $\dot{\Sigma}[Z] = \begin{pmatrix} 2Z_S + Z_a I_q & \Lambda Z_C^\top \\ Z_C \Lambda & Z_a I_{p-q} \end{pmatrix}$, and the entrywise asymptotic variances compare with those of S_n as follows:*

entry	$\widehat{\Sigma}_n$	S_n
$(U, U), (i, i)$	$2\lambda_i^2$	$2\lambda_i^2$
$(U, U), (i, j), i \neq j$	$\lambda_i \lambda_j$	$\lambda_i \lambda_j$
$(U_\perp, U), (k, i)$	$\sigma_0^2 \lambda_i$	$\sigma_0^2 \lambda_i$
$(U_\perp, U_\perp), (k, k)$	$2\sigma_0^4/(p-q)$	$2\sigma_0^4$
$(U_\perp, U_\perp), (k, l), k \neq l$	0	σ_0^4

Moreover $\text{Cov}((U, U)_{ii} \text{ error}, \text{noise-block error}) = \text{Cov}(2Z_{S_{ii}} + Z_a, Z_a) = 0$.

Proposition 5.7 (Efficiency dominance over the sample covariance). *Let $\mathcal{F}(T, T') := \frac{1}{2} \text{tr}(\Sigma_0^{-1} T \Sigma_0^{-1} T')$ be the Fisher metric of the unrestricted Gaussian covariance model on $\text{Sym}(p)$, with operator form $\mathcal{F}T = \frac{1}{2} \Sigma_0^{-1} T \Sigma_0^{-1}$, $\mathcal{F}^{-1}T = 2\Sigma_0 T \Sigma_0$. Then $\text{Avar}(S_n) = \mathcal{F}^{-1}$, $G = i^* \mathcal{F} i$, $P := i G^{-1} i^* \mathcal{F}$ is the $\mathcal{F}$ -orthogonal projector onto $\text{range}(i)$, and*

$$\text{Avar}(\widehat{\Sigma}_n) = P \mathcal{F}^{-1} = P \mathcal{F}^{-1} P^*, \quad \mathcal{F}^{-1} - \text{Avar}(\widehat{\Sigma}_n) = (I - P) \mathcal{F}^{-1} (I - P)^* \succeq 0.$$

Remark 5.8 (No eigengap). The classical route — eigenvector perturbation of S_n plus the delta method [1] — needs the leading q eigenvalues of Σ_0 simple. Theorems 4.2–5.4 need nothing of the sort: with repeated singular values individual principal directions are not asymptotically normal, but the point $[\theta] \in \mathcal{Q}$ is. This mirrors the conceptual gain [13] demonstrate for single-index models (§5.2 there).

6 Proofs

Step 0 (Localization). By Datta’s Corollaries 1–2 and Lemma 2.9, $d_{\mathcal{Q}}([\widehat{\theta}_n], [\theta_0]) \rightarrow 0$ in probability (a.s. in the i.i.d. case). Hence $A_n := \{[\widehat{\theta}_n] \in B([\theta_0], r_0)\}$ has $\mathbb{P}(A_n) \rightarrow 1$; on A_n , Proposition 2.8 makes $\widehat{u}_n := \text{Exp}_{[\theta_0]}^{-1}[\widehat{\theta}_n] = e_n$ well defined, and by Lemma 3.3 the alignment defining e_n is a.s. unique for every $n \geq p$. All o_P -statements below are on A_n (or after the modification $\widehat{u}_n \mathbf{1}_{A_n}$, which changes nothing asymptotically by the argument in Proposition 3.5).

Step 1 (Proof of Theorem 5.2). *Reduction to a Euclidean family.* By Proposition 2.5, the manifold perturbation is exactly a linear perturbation of the lift:

$$P_{\text{Exp}_{[\theta_0]}(h)} = P_h, \quad \frac{dP_h}{dx}(x) = f(x; \theta_0 + \bar{h}) = (2\pi)^{-p/2} \det \Sigma(h)^{-1/2} \exp\left(-\frac{1}{2} x^\top \Sigma(h)^{-1} x\right),$$

where $\Sigma(h) := \Sigma(\theta_0 + \bar{h})$. Hence $\{P_h : h \in B_{\mathcal{H}}(0, r_0)\}$ is an ordinary parametric family indexed by an open subset of the Euclidean space $\mathcal{H}_{\theta_0} \cong \mathbb{R}^{\dim \mathcal{Q}}$, and Definition 3.7 at $[\theta_0]$ is classical DQM of this family at $h = 0$ — no approximation of the exponential map is involved.

Verification of van der Vaart’s criterion [15, Lemma 7.6] (“if for every x the map $h \mapsto f(x; h)$ is continuously differentiable and the Fisher information matrix $I(h) = \mathbb{E}_h[\dot{\ell}_h \dot{\ell}_h^\top]$ is well defined and continuous, then the family is DQM with score $\dot{\ell}_h$ ”):

- (a) *Smoothness of the densities.* $h \mapsto \Sigma(h) = \Sigma_0 + \dot{\Sigma}[\bar{h}] + \bar{h}_W \bar{h}_W^\top$ is a quadratic polynomial with values, on the ball, in the compact set $\{\Sigma : \frac{\sigma_0^2}{2} I \preceq \Sigma \preceq c_2 I\}$ of positive-definite matrices (Weyl; $r_0 < \sigma_0^2/2$ after a final shrinkage), where $c_2 := (\ell_1 + r_0)^2 + \sigma_0^2 + r_0$. On the positive-definite cone, $\Sigma \mapsto \Sigma^{-1}$ and $\Sigma \mapsto \log \det \Sigma$ are real-analytic, so $h \mapsto f(x; h)$ is C^∞ for every x , with score obtained from $d \log \det \Sigma = \text{tr}(\Sigma^{-1} d\Sigma)$ and $d(\Sigma^{-1}) = -\Sigma^{-1}(d\Sigma)\Sigma^{-1}$:

$$\dot{\ell}_h[v] = -\frac{1}{2} \text{tr}(\Sigma(h)^{-1} \dot{\Sigma}_h[\bar{v}]) + \frac{1}{2} x^\top \Sigma(h)^{-1} \dot{\Sigma}_h[\bar{v}] \Sigma(h)^{-1} x = \frac{1}{2} \text{tr}(\Sigma(h)^{-1} (xx^\top - \Sigma(h)) \Sigma(h)^{-1} \dot{\Sigma}_h[\bar{v}]), \quad (17)$$

where $\dot{\Sigma}_h[\bar{v}] = \bar{v}_W(W_0 + \bar{h}_W)^\top + (W_0 + \bar{h}_W)\bar{v}_W^\top + v_a I_p$ is the differential of Σ at $\theta_0 + \bar{h}$, affine in h .

- (b) *Fisher matrix, finite and continuous.* By Lemma A.1 applied at $\Sigma(h)$ exactly as in Lemma 4.1, $I(h)(v, v') = \frac{1}{2} \text{tr}(\Sigma(h)^{-1} \dot{\Sigma}_h[\bar{v}] \Sigma(h)^{-1} \dot{\Sigma}_h[\bar{v}'])$, a composition of continuous maps of h (inverse on the compact positive-definite set; affine $\dot{\Sigma}_h$), hence finite and continuous on the ball.

Lemma 7.6 yields (9) at $h = 0$ with score (17) at $h = 0$, i.e. (10), a bounded linear functional of h , hence an element of $T_{[\theta_0]}^* \mathcal{Q}$ under the isometric identification $d\pi$. Only rank $W = q$, $q < p$, $\sigma^2 > 0$ were used, so the same argument applies at every base point $[\theta] \in \mathcal{Q}$.

LAN and contiguity. DQM at $[\theta_0]$ is the sole hypothesis of [13, Prop. 3.1], which gives (16); the classical CLT applied to the i.i.d. variables $s(x_i)(h)$ (mean zero, variance $G(h, h)$ by Lemma 4.1) then shows the log-likelihood ratio converges weakly, under P_{θ_0} , to $\mathcal{N}(-\frac{1}{2}G(h, h), G(h, h))$. A limit of the form $\mathcal{N}(-\frac{1}{2}\tau^2, \tau^2)$ has $\mathbb{E}e^\mathcal{N} = 1$, so by Le Cam's first lemma ([15, Example 6.5]) the two product sequences are mutually contiguous. $\square$

Step 2 (Proof of Theorem 5.3). Work in the normal coordinates $u = \text{Exp}_{[\theta_0]}^{-1}(\cdot)$ of Proposition 2.8, i.e. with the Euclidean family $\{P_u\}$ of Step 1, and set $\ell(u; x) := \log f(x; \theta_0 + \bar{u})$. On A_n the global maximizer $[\hat{\theta}_n]$ of the likelihood over $\mathcal{Q}$ lies in the chart image, so $\hat{u}_n$ maximizes $u \mapsto \sum_i \ell(u; x_i)$ over the ball $\bar{B}_{\mathcal{H}}(0, r_0)$; off A_n redefine $\hat{u}_n := 0$ (probability $\rightarrow 0$). We verify the four hypotheses of [15, Thm. 5.23] for $m_u := \ell(u; \cdot)$ at the population maximizer $u = 0$.

(a) *Measurability and differentiability at 0.* For each x , $u \mapsto \ell(u; x)$ is C^∞ on the ball (Step 1(a)); the derivative at 0 is the score $s(x) \in L^2(P_{\theta_0})$ (Lemma 4.1).

(b) *Square-integrable Lipschitz envelope on the ball.* On $\|u\| \leq r_0$ the eigenvalues of $\Sigma(u)$ lie in $[c_1, c_2]$, $c_1 := \sigma_0^2/2$, and

$$\|\dot{\Sigma}_u[\bar{v}]\|_F \leq 2\|W_0 + \bar{u}_W\|_2 \|\bar{v}_W\|_F + |v_a| \sqrt{p} \leq (2(\ell_1 + r_0) + \sqrt{p}) \|v\| =: c_3 \|v\|.$$

From (17), using $|\text{tr}(MN)| \leq \|M\|_F \|N\|_F$, $\|\Sigma(u)^{-1}\|_2 \leq c_1^{-1}$ and $\|xx^\top - \Sigma(u)\|_F \leq \|x\|^2 + \sqrt{p} c_2$,

$$|\nabla_u \ell(u; x)[v]| \leq \frac{1}{2} c_1^{-2} (\|x\|^2 + \sqrt{p} c_2) c_3 \|v\| =: m(x) \|v\|, \quad \mathbb{E} m(X)^2 < \infty$$

(Gaussian fourth moments). Since the ball is convex and ℓ is C^1 in u , the mean value theorem gives $|\ell(u_1; x) - \ell(u_2; x)| \leq m(x) \|u_1 - u_2\|$ for all u_1, u_2 in the ball.

(c) *Second-order expansion of the population objective with nonsingular Hessian.* The model is correctly specified, so

$$\mathbb{E}_{\theta_0} \ell(u; X) - \mathbb{E}_{\theta_0} \ell(0; X) = \mathbb{E}_{\theta_0} \log \frac{f(X; \theta_0 + \bar{u})}{f(X; \theta_0)} = -\text{KL}(P_0 \| P_u) = -\frac{1}{2} G(u, u) + o(\|u\|^2)$$

by Lemma A.2 (where the $\ddot{\Sigma}$ -terms cancel identically, and the ray-wise expansion upgrades to a genuine Taylor expansion since $u \mapsto \text{KL}$ is C^∞ near 0, being a composition of the polynomial $\Sigma(\cdot)$)

with real-analytic functions on the PD cone). The Hessian $-G$ is negative definite by Theorem 4.2, and $u = 0$ maximizes $(\text{KL} \geq 0$ with equality only at $\Sigma(u) = \Sigma_0$, i.e. $u = 0$ by injectivity).

(d) *Consistency and near-maximization.* $\hat{u}_n \rightarrow_P 0$ by Step 0; on A_n , $\hat{u}_n$ is an exact maximizer over the ball.

[15, Thm. 5.23] yields $\sqrt{n} \hat{u}_n = G^{-1} n^{-1/2} \sum_i s(x_i) + o_{P_{\theta_0}}(1)$, and the multivariate CLT for the i.i.d. scores gives $\sqrt{n} \hat{u}_n \rightsquigarrow \mathcal{N}(0, G^{-1} G G^{-1}) = \mathcal{N}(0, G^{-1})$. Since $\hat{u}_n$ is $\text{Exp}_{[\theta_0]}^{-1}[\hat{\theta}_n]$ by definition, this is verbatim the display of Theorem 5.3. $\square$

Step 3 (Regularity: proof of Theorem 5.4(i)). Fix h , set $u_n := h/\sqrt{n}$ and $P_n := P_{[\theta]_{n,h}}^{\otimes n}$, $P_0 := P_{[\theta_0]}^{\otimes n}$.

(i) *Coordinate regularity via Le Cam's third lemma.* Fix any $h' \in \mathcal{H}_{\theta_0}$ and set $T_n(h') := n^{-1/2} \sum_i s(x_i)(h')$, $\Lambda_n := \log dP_n/dP_0$. By (16), $\Lambda_n = T_n(h) - \frac{1}{2}G(h, h) + o_{P_0}(1)$, and the bivariate CLT for the i.i.d. pairs $(s(x_i)(h'), s(x_i)(h))$ (mean zero, covariance matrix $\begin{pmatrix} G(h', h') & G(h', h) \\ G(h, h') & G(h, h) \end{pmatrix}$) by Lemma 4.1) gives

$$(T_n(h'), \Lambda_n) \rightsquigarrow_{P_0} \left(N_1, N_2 - \frac{1}{2}G(h, h) \right), \quad \text{Cov}(N_1, N_2) = G(h', h).$$

Le Cam's third lemma ([15, Thm. 6.6 and Example 6.7]) yields $T_n(h') \rightsquigarrow_{P_n} \mathcal{N}(G(h', h), G(h', h'))$; since h' was arbitrary and the dimension is finite, the score vector satisfies $n^{-1/2} \sum_i s(x_i) \rightsquigarrow_{P_n} \mathcal{N}(Gh, G)$. The linear expansion of Step 2 has remainder $o_{P_0}(1)$, hence $o_{P_n}(1)$ by contiguity (Theorem 5.2; convergence in probability to a constant transfers along contiguous sequences), and the same transfer gives $\|\hat{u}_n\| \rightarrow_{P_n} 0$. Therefore

$$\sqrt{n} \hat{u}_n \rightsquigarrow_{P_n} \mathcal{N}(h, G^{-1}), \quad \text{i.e.} \quad \sqrt{n} (\hat{u}_n - u_n) \rightsquigarrow_{P_n} \mathcal{N}(0, G^{-1}). \quad (18)$$

(ii) *Transport comparison.* It remains to replace the coordinate difference by the intrinsic transported error.

Lemma 6.1 (Transport comparison). *Define, for $u, w \in \bar{B}_{\mathcal{H}}(0, r_0/2)$,*

$$F(u, w) := \Pi_{\text{Exp}_{[\theta_0]}(u)}^{[\theta_0]} \text{Exp}_{\text{Exp}_{[\theta_0]}(u)}^{-1}(\text{Exp}_{[\theta_0]}(w)) \in T_{[\theta_0]} \mathcal{Q}.$$

Then F is C^∞ on the compact product ball, and there is $C = C(r_0)$ with

$$\|F(u, w) - (w - u)\| \leq C \|u\| \|w - u\| \quad \text{for all such } u, w.$$

Proof. Smoothness. Write $\theta(u) := \theta_0 + \bar{u}$, $\theta(w) := \theta_0 + \bar{w}$. By Lemma 2.7 and Proposition 2.8 applied at the base point $\theta(u)$, the logarithm is the aligned difference

$$\text{Exp}_{\text{Exp}(u)}^{-1}(\text{Exp}(w)) = \left((W_0 + \bar{w}_W) \hat{R}(u, w) - (W_0 + \bar{u}_W), w_a - u_a \right), \quad \hat{R}(u, w) = M (M^\top M)^{-1/2},$$

$M := (W_0 + \bar{w}_W)^\top (W_0 + \bar{u}_W)$, using that the polar factor of an invertible matrix (M is invertible on the ball, by the proof of Proposition 2.8(a) at the base $\theta(u)$) is the smooth function $M(M^\top M)^{-1/2}$ of M , itself affine in (u, w) . The transport $\Pi_{\text{Exp}(u)}^{[\theta_0]}$ acts along the minimizing geodesic $t \mapsto \text{Exp}_{[\theta_0]}(tu)$ and is the solution map of the linear ODE (6), whose coefficients are smooth in (t, u) with $\sigma_{\min}(W_0 + t\bar{u}_W)$ bounded below on the ball; by smooth dependence of linear ODEs on parameters, Π is smooth in $(u, \text{input vector})$ and linear in the input. Hence $F \in C^\infty$ on the compact product ball, with all derivatives bounded.

The bound. Two identities: $F(0, w) = w$ for all w (the geodesic from $[\theta_0]$ to itself is trivial and the transport is the identity; and $\text{Exp}_{[\theta_0]}^{-1} \text{Exp}_{[\theta_0]}(w) = w$ on the ball by Proposition 2.8); and

$F(u, u) = 0$ for all u ($\text{Exp}_x^{-1}(x) = 0$, transported to 0). Let $H(u, w) := F(u, w) - (w - u)$; then $H(0, \cdot) \equiv 0$ and $H(u, u) = 0$. Because $H(0, \cdot) \equiv 0$, its partial derivative satisfies $\partial_w H(0, w) = 0$ for every w ; and $\partial_w H$ is Lipschitz on the compact ball with some constant L (bounded second derivatives), so $\|\partial_w H(u, v)\| \leq L\|u\|$ for all v in the ball. Therefore

$$\|H(u, w)\| = \left\| \int_0^1 \partial_w H(u, u + t(w - u)) [w - u] dt \right\| \leq L \|u\| \|w - u\|. \quad \square$$

(iii) *Assembly.* Apply Lemma 6.1 with $u = u_n$ ($\|u_n\| = \|h\|/\sqrt{n}$, eventually in the ball) and $w = \hat{u}_n$ (in the ball with P_n -probability $\rightarrow 1$ by the transfer above):

$$\begin{aligned} \left\| \Pi_{[\theta]_{n,h}}^{[\theta_0]} \sqrt{n} \text{Exp}_{[\theta]_{n,h}}^{-1}[\hat{\theta}_n] - \sqrt{n}(\hat{u}_n - u_n) \right\| &= \sqrt{n} \|H(u_n, \hat{u}_n)\| \\ &\leq C\sqrt{n} \frac{\|h\|}{\sqrt{n}} \|\hat{u}_n - u_n\| = C\|h\| \cdot O_{P_n}(n^{-1/2}) \rightarrow 0. \end{aligned}$$

With (18) this gives Definition 3.9 with $Z_{[\theta_0]} \sim \mathcal{N}(0, G^{-1})$, whose law is free of h . $\square$

Step 4 (Optimality: proof of Theorem 5.4(ii)–(iii)). (ii) *Convolution.* The hypotheses of [13, Thm. 3.1] are: the model is DQM at $[\theta_0]$ (Theorem 5.2); the parameter map ψ satisfies their Assumption 3.1 (here $\psi = \text{id}$, $\dot{\psi} = \text{id}$); the estimator sequence is regular with some limit law L (Step 3, with $L = \mathcal{N}(0, G^{-1})$ by Theorem 5.3 for the MLE). Their conclusion is $L = \mathcal{N}(0, \dot{\psi}G^{-1}\dot{\psi}^*) * \Delta = \mathcal{N}(0, G^{-1}) * \Delta$ for every regular sequence, and the MLE's limit is the pure Gaussian factor.

(iii) *LAM.* We verify the hypotheses of [13, Lem. 3.3 and Thm. 3.3], using their own prior construction (proof of their Theorem 3.3): priors $\pi_{\theta_0, c/\sqrt{n}}$ obtained by pushing a sufficiently regular density H , compactly supported on the unit ball of $T_{[\theta_0]}\mathcal{Q}$, through $h \mapsto \text{Exp}_{[\theta_0]}(ch/\sqrt{n})$ — supported in the normal ball for n large. Then: $G_\pi \succ 0$ for such H (their construction); $\psi = \text{id}$ is continuously differentiable; the prior densities are continuous and compactly supported; the model has measurable densities with respect to Lebesgue measure; DQM holds at every point of the support (Theorem 5.2 at all base points); and $\gamma(\pi)$, $\mathbb{E}_\pi[\dot{\psi}]$ and $\int G d\pi$ are finite because $[\theta] \mapsto G_{[\theta]}$ is continuous — in the chart, $G(u)(v, v') = \frac{1}{2} \text{tr}(\Sigma(u)^{-1} \dot{\Sigma}_u[v] \Sigma(u)^{-1} \dot{\Sigma}_u[v'])$ is continuous in u (Step 1(b)) — and the support is compact. Their Assumption C.2 holds by their own sufficient condition: $\dot{\psi}$ and G continuous at $[\theta_0]$ with $G_{[\theta_0]}$ nonsingular (Theorem 4.2). [13, Thm. 3.3] then delivers the LAM bound with covariance $\dot{\psi}G^{-1}\dot{\psi}^* = G^{-1}$, attained by the MLE (Theorem 5.3).

Uniqueness of the influence operator. By [13, Thm. 3.2], any regular asymptotically linear sequence satisfies $\mathbb{E}[\text{IF} \cdot S(h)] = h$ for all h ; in this parametric model the components of IF lie in the finite-dimensional score space $\Lambda = \{s(\cdot)(h)\}$ (after L^2 -projection, which changes neither the expansion nor the limit), and on Λ the system $\mathbb{E}[\text{IF} \cdot S(h)] = h$ has the unique solution $\text{IF} = G^{-1}s$. $\square$

Step 5 (Proof of Corollaries 5.5, 5.6 and Proposition 5.7). *Corollary 5.5.* The expansion $\Sigma(\theta_0 + \bar{u}) - \Sigma_0 = \dot{\Sigma}[\bar{u}] + \bar{u}_W \bar{u}_W^\top$ is *exact*, so on A_n

$$\sqrt{n}(\hat{\Sigma}_n - \Sigma_0) = i(\sqrt{n}\hat{u}_n) + \sqrt{n}\bar{u}_{W,n}\bar{u}_{W,n}^\top, \quad \|\sqrt{n}\bar{u}_{W,n}\bar{u}_{W,n}^\top\|_F \leq \sqrt{n}\|\hat{u}_n\|^2 = O_P(n^{-1/2}),$$

and Theorem 5.3 with the continuous mapping theorem gives the CLT with covariance $iG^{-1}i^*$. The alternative form of $\hat{\Sigma}_n$ is Tipping–Bishop algebra: $\widehat{W}\widehat{W}^\top + \hat{\sigma}^2 I = U_q(\Delta_q - \hat{\sigma}^2 I)U_q^\top + \hat{\sigma}^2 I = U_q\Delta_q U_q^\top + \hat{\sigma}^2(I - U_q U_q^\top)$.

Corollary 5.6. $\dot{\Sigma}[Z]$ has the displayed blocks by (12) with (S, C, a) replaced by (Z_S, Z_C, Z_a) . The variances are read off Proposition 4.5: $(U, U)_{ii}$ error $= 2Z_{S_{ii}} + Z_a$ with variance $2\lambda_i^2$ (computation

in Remark 4.9(2)); $(U, U)_{ij}$ error = $2Z_{S_{ij}}$ with variance $4 \cdot \lambda_i \lambda_j / 4$; $(U_\perp, U)_{ki}$ error = $\ell_i Z_{C_{ki}}$ with variance $\ell_i^2 \cdot \sigma_0^2 \lambda_i / \ell_i^2$; noise-block errors $Z_a \delta_{kl}$. For S_n : with $y = U^\top x$, $z = U_\perp^\top x$ independent, $\text{Var}(y_i^2) = 2\lambda_i^2$, $\text{Var}(y_i y_j) = \lambda_i \lambda_j$, $\text{Var}(z_k y_i) = \sigma_0^2 \lambda_i$, $\text{Var}(z_k^2) = 2\sigma_0^4$, $\text{Var}(z_k z_l) = \sigma_0^4$. The covariance statement: $\text{Cov}(2Z_{S_{ii}} + Z_a, Z_a) = 2(-\sigma_0^4/(p-q)) + 2\sigma_0^4/(p-q) = 0$.

Proposition 5.7. For $T \in \text{Sym}(p)$, $\text{Var}(x^\top T x) = 2 \text{tr}((T \Sigma_0)^2) = \langle T, \mathcal{F}^{-1} T \rangle$ by Lemma A.1, and $n \text{Var}(\langle T, S_n \rangle) = \text{Var}(x^\top T x)$, so $\text{Avar}(S_n) = \mathcal{F}^{-1}$ as a quadratic form. $G = i^* \mathcal{F} i$ is (11) read through $\mathcal{F}$. Set $P := iG^{-1}i^* \mathcal{F}$. Then $P^2 = iG^{-1}(i^* \mathcal{F} i)G^{-1}i^* \mathcal{F} = iG^{-1}i^* \mathcal{F} = P$; $\mathcal{F}P = \mathcal{F}iG^{-1}i^* \mathcal{F}$ is self-adjoint, so $P^* = \mathcal{F}P\mathcal{F}^{-1}$, i.e. P is $\mathcal{F}$ -self-adjoint and hence the $\mathcal{F}$ -orthogonal projector onto $\text{range}(i)$. Consequently $P\mathcal{F}^{-1} = iG^{-1}i^* = \text{Avar}(\widehat{\Sigma}_n)$; $(P\mathcal{F}^{-1})^* = \mathcal{F}^{-1}P^* = \mathcal{F}^{-1}\mathcal{F}P\mathcal{F}^{-1} = P\mathcal{F}^{-1}$, so it is self-adjoint, and $P\mathcal{F}^{-1}P^* = P\mathcal{F}^{-1}\mathcal{F}P\mathcal{F}^{-1} = P^2\mathcal{F}^{-1} = P\mathcal{F}^{-1}$. Finally

$$\begin{aligned} (I - P)\mathcal{F}^{-1}(I - P)^* &= (I - P)\mathcal{F}^{-1} - (I - P)\mathcal{F}^{-1}P^* = (I - P)\mathcal{F}^{-1} - (I - P)P\mathcal{F}^{-1} \\ &= (I - P)\mathcal{F}^{-1} = \mathcal{F}^{-1} - P\mathcal{F}^{-1}, \end{aligned}$$

which is of the form $M\mathcal{F}^{-1}M^* \succeq 0$ since $\mathcal{F}^{-1} \succ 0$. $\square$

7 Consequences: what the theorem says and how to use it

Theorem 5.3 asserts the weak convergence of a sequence of *random tangent vectors* $\sqrt{n} \text{Exp}_{[\theta_0]}^{-1}[\widehat{\theta}_n] \in T_{[\theta_0]}\mathcal{Q}$ to a Gaussian law on that tangent space. This section explains what such a statement means geometrically, and what an applied statistician who fits a PPCA model can *do* with it.

7.1 The geometric picture

The tangent space is the linearization of the model at the truth. $\mathcal{Q}$ is curved and has no vector-space structure: one cannot subtract $[\widehat{\theta}_n]$ from $[\theta_0]$. But at the single point $[\theta_0]$ there is a genuine Euclidean space, $T_{[\theta_0]}\mathcal{Q}$, of dimension $pq + 1 - \frac{q(q-1)}{2}$; the logarithm $\text{Exp}_{[\theta_0]}^{-1}$ is the canonical dictionary that turns a point of $\mathcal{Q}$ near $[\theta_0]$ into a vector in that space — the initial velocity of the unique shortest geodesic pointing from the truth to the estimate. Its length is the distance and its direction is the direction. Asymptotic normality says: *these arrows, rescaled by $\sqrt{n}$, fill out a Gaussian cloud.*

Three readings of the same picture.

- (i) *Arrows at the truth.* Repeat the experiment many times. Each replicate yields an estimate $[\widehat{\theta}_n]$; draw the geodesic from $[\theta_0]$ to it and record its initial velocity, magnified by $\sqrt{n}$. The resulting cloud of arrows in $T_{[\theta_0]}\mathcal{Q}$ is asymptotically an ellipsoidal Gaussian with covariance G^{-1} — with *no* spread in the $\frac{q(q-1)}{2}$ vertical (rotational) directions, which are simply not present in the tangent space.
- (ii) *Procrustes-aligned parameters.* Concretely (Proposition 3.5), the arrow is $(\widehat{W}_n \widehat{R}_n - W_0, \widehat{\sigma}_n^2 - \sigma_0^2)$: rotate each fitted loading matrix optimally onto the truth, then subtract. Alignment is not a cosmetic post-processing step; it *is* the intrinsic subtraction.
- (iii) *Curvature is a second-order effect.* A geodesic ball of radius ϵ in $\mathcal{Q}$ differs from a Euclidean ball of the same radius only at order ϵ^3 in volume (Bishop–Günther), and the estimator is inside a ball of radius $O_P(n^{-1/2})$. The curvature bound $\kappa(r_0)$ of Proposition 2.11 is what makes the transport comparison (Lemma 6.1) an $O(\|u\|\|w - u\|)$ error; at the $\sqrt{n}$ scale it

washes out. So: *first-order inference on a curved parameter space is Euclidean inference in the tangent space at the truth* — curvature would only surface in higher-order (Edgeworth-type) refinements.

What is *not* being claimed. Not that $\widehat{W}_n$ is asymptotically normal — it is not; without alignment it wanders on the orbit. Not that the entries of $\widehat{W}_n$ have a nondegenerate joint Gaussian limit in $\mathbb{R}^{pq+1}$ — the limit is supported on a proper subspace, $\mathcal{H}_{\theta_0}$, of codimension $\frac{q(q-1)}{2}$. Not that individual eigenvectors are normal — they are not when eigenvalues tie (Remark 5.8); it is the *class* that is normal.

7.2 What the applied statistician gains

Suppose one has fitted PPCA with q components to n observations, obtaining $(\widehat{W}_n, \widehat{\sigma}_n^2)$, and wants to report uncertainty.

Recipe 7.1 (Standard errors from the plug-in Fisher operator). Compute the SVD $\widehat{W}_n = \widehat{U}\widehat{\Lambda}\widehat{V}^\top$, set $\widehat{\lambda}_i = \widehat{\ell}_i^2 + \widehat{\sigma}_n^2$, and read off closed-form standard errors from Proposition 4.5 with hatted quantities:

$$\text{SE}(\widehat{\sigma}_n^2) = \sqrt{\frac{2\widehat{\sigma}_n^4}{n(p-q)}}, \quad \text{SE}(k\text{-th noise-direction tilt of } \widehat{u}_i) = \sqrt{\frac{\widehat{\sigma}_n^2 \widehat{\lambda}_i}{n\widehat{\ell}_i^4}},$$

the latter from (15) divided by $\widehat{\ell}_i$. No bootstrapping, no numerical Hessian.

Remark 7.2 (The noise variance is estimated much better than one might think). Corollary 4.7 says $\text{Var}(\widehat{\sigma}_n^2) \approx 2\sigma_0^4/(n(p-q))$: the accuracy improves *linearly in the number of discarded dimensions*. With $p = 100$, $q = 5$, the noise level is pinned down as if one had $95n$ observations of a scalar variance. The influence function $\text{IF}_{\sigma^2}(x) = (\|U_\perp^\top x\|^2 - (p-q)\sigma_0^2)/(p-q)$ makes this transparent and also tells the analyst *which observations drive* $\widehat{\sigma}_n^2$: those with large residual norm off the fitted subspace — an immediate influence diagnostic (plot $\text{IF}_{\sigma^2}(x_i)$ against i).

Recipe 7.3 (Wald tests and confidence regions with the right degrees of freedom). Let $\widehat{G}$ be the plug-in Fisher operator (Proposition 4.4) as a matrix in the coordinates of Lemma 4.3. For a hypothesised $[\theta^\dagger]$ near the estimate,

$$T_n := n \langle \widehat{G} \text{Exp}_{[\widehat{\theta}_n]}^{-1}[\theta^\dagger], \text{Exp}_{[\widehat{\theta}_n]}^{-1}[\theta^\dagger] \rangle \rightsquigarrow \chi_d^2, \quad d = \dim \mathcal{Q} = pq + 1 - \frac{q(q-1)}{2},$$

where the logarithm is computed by Procrustes alignment (Proposition 2.8). *The degrees of freedom are the crux*: an analyst who naively treats (W, σ^2) as a free $(pq+1)$ -vector inflates d by $\frac{q(q-1)}{2}$ — e.g. by 10 when $q = 5$ — and gets systematically conservative tests. The same count is the correct one for likelihood-ratio tests, and it reconciles with the familiar statement that a PPCA fit has $pq + 1 - q(q-1)/2$ free parameters.

Recipe 7.4 (Confidence regions are exponential images of ellipsoids). A $(1-\alpha)$ confidence region for $[\theta_0]$ is

$$\mathcal{R}_{n,\alpha} = \text{Exp}_{[\widehat{\theta}_n]} \left(\left\{ h \in T_{[\widehat{\theta}_n]} \mathcal{Q} : n \langle \widehat{G}h, h \rangle \leq \chi_{d,1-\alpha}^2 \right\} \right),$$

i.e. take the Wald ellipsoid in the tangent space at the estimate and push it onto the manifold along geodesics — concretely, by Proposition 2.5, $h \mapsto [(\widehat{W}_n, \widehat{\sigma}_n^2) + \bar{h}]$ with $\bar{h}$ horizontal at $\widehat{W}_n$. Every parameter in the region is a legitimate PPCA parameter (positive σ^2 , full-rank W); nothing needs

to be projected back afterwards. Contrast with a naive Euclidean Wald region in (W, σ^2) , which contains rotational directions along which the likelihood is exactly flat, and hence is infinitely wide in those directions unless one first quotients them out.

Recipe 7.5 (Aligned bootstrap). If one prefers resampling, the theorem says exactly what to resample: draw bootstrap samples, refit to get $\widehat{W}_n^{(b)}$, align $\widehat{R}^{(b)} = \arg \min_R \|\widehat{W}_n^{(b)} R - \widehat{W}_n\|_F$, and treat $\{(\widehat{W}_n^{(b)} \widehat{R}^{(b)} - \widehat{W}_n, \widehat{\sigma}^{2,(b)} - \widehat{\sigma}_n^2)\}_b$ as draws from the sampling distribution in $T_{[\widehat{\theta}_n]} \mathcal{Q}$. *Without* the alignment step, bootstrap loading matrices are incomparable across replicates and the resulting “standard errors” are meaningless (they measure orbit motion, not statistical error). The $\sqrt{n}$ -consistency and regularity established here are precisely the conditions under which this aligned bootstrap is asymptotically valid.

Which functionals are safe to report. Anything that is a smooth function of $[\theta]$ inherits a CLT and an efficiency bound by the delta method (Remark 3.6): the noise variance σ^2 ; the principal *subspace* $\text{col}(W)$ as a point of the Grassmannian; the spectrum $\{\ell_i\}$; explained-variance ratios; the fitted covariance $\Sigma(\theta)$ (Corollary 5.5). Anything that is *not* a function of $[\theta]$ — an individual loading entry W_{ji} , the sign or ordering of a component, an individual eigenvector when eigenvalues tie — has no well-defined sampling distribution, and any standard error reported for it is an artifact of the software’s arbitrary rotation convention. The quotient viewpoint draws this line sharply, and Theorem 4.2 explains why the line falls where it does: $\ker \dot{\Sigma}$ is exactly the rotational directions.

Efficiency has a concrete payoff. Theorem 5.4 says no regular estimator beats the MLE, and Proposition 5.7 localizes the gain relative to the unrestricted sample covariance: by Corollary 5.6, $\widehat{\Sigma}_n$ and S_n have identical first-order accuracy in the signal block and the cross block, but the noise block improves by a factor $p - q$ (diagonal) and collapses to zero variance (off-diagonal). In words: *imposing the PPCA structure buys nothing where the data speak directly (the leading subspace) and buys a great deal where the model imposes structure (the isotropic noise floor)*. That is also a warning — the gain is only real if the isotropy assumption is; the same influence function IF_{σ^2} furnishes the natural diagnostic, since heteroscedastic noise shows up as systematic structure in the residual norms $\|U_{\perp}^{\top} x_i\|^2$.

8 State of the argument

Complete in this draft: the geometric toolkit of Section 2 (Lemmas 2.1–2.9, Propositions 2.5, 2.8, 2.10, 2.11), including the local distance identity, the geodesic property with an explicit radius, uniqueness of minimizing geodesics by chord equality, and the curvature bound; the almost-sure alignment uniqueness (Lemma 3.3); Theorem 4.2; the DQM/LAN Theorem 5.2 with full verification of van der Vaart’s Lemma 7.6; the explicit Fisher calculus of Section 4 with a moment-identity verification and three classical cross-checks, plus the independent Davis–Kahan derivation of IF_{σ^2} (Lemma 4.8); Theorem 5.3 with all four hypotheses of [15, Thm. 5.23] verified; the regularity proof, including Le Cam’s third lemma and the transport-comparison Lemma 6.1; the optimality Step 4, with the hypotheses of [13, Lem. 3.3, Thm. 3.3] checked one by one; Corollaries 5.5, 5.6 and Proposition 5.7; the appendix identities.

Still to be supplied: (a) explicit numerical constants in Lemma 2.7 (the compactness argument is complete but not quantitative); (b) the dependent-data variant (Section 9); (c) a simulation

appendix (QQ-plots of the coordinates of $\sqrt{n} \text{Exp}_{[\theta_0]}^{-1}[\hat{\theta}_n]$ against $\mathcal{N}(0, G^{-1})$, and coverage of the regions of Recipe 7.4); (d) the extension to $\text{rank}(W_0) < q$ (a singular case, excluded here as in [13]).

9 Extensions

1. *Rates.* Theorem 5.3 gives $d_Q([\hat{\theta}_n], [\theta_0]) = O_P(n^{-1/2})$ — the “rates” half of the thesis’s open question, as a byproduct.
2. *Dependent data.* Combine Datta’s Theorems 7–9 with a LAN condition under mixing ([13, Rem. 3.4]); the limit acquires a long-run-variance sandwich $G^{-1}\mathcal{V}G^{-1}$, and Corollary 4.7 generalizes with the long-run variance of $\|U_\perp^\top x\|^2$.
3. *Misspecification / elliptical noise.* The same geometry with the sandwich $(\nabla^2 M)^{-1}\mathbb{E}[\phi \otimes \phi](\nabla^2 M)^{-1}$ of [13, Prop. 5.1]; the explicit blocks of Proposition 4.4 make the kurtosis corrections computable.
4. *Functionals.* $\text{col}(W) \in \text{Gr}(p, q)$, explained variance, the spectrum: smooth functionals of $[\theta]$, with limit laws and efficiency bounds by the manifold delta method; (15) is the key input for the subspace functional.
5. *Missing data; Bernstein–von Mises.* AIPW-type efficient estimation of $[\theta]$ under MAR via [13, Rem. 5.3]; a BvM theorem on $\mathcal{Q}$ using the LAN expansion (16) (a sequel to Chapters 2–3 of the thesis).

A Gaussian identities

Lemma A.1 (Quadratic-form covariance). *For $x \sim \mathcal{N}(0, \Sigma)$ and $M, N \in \text{Sym}(p)$,*

$$\text{Cov}(x^\top Mx, x^\top Nx) = 2 \text{tr}(M\Sigma N\Sigma).$$

Proof. Whitening $x = \Sigma^{1/2}z$, $z \sim \mathcal{N}(0, I)$, reduces the claim to $\widetilde{M} = \Sigma^{1/2}M\Sigma^{1/2}$ and $\widetilde{N} = \Sigma^{1/2}N\Sigma^{1/2}$. By Isserlis’ theorem, $\mathbb{E}[z_i z_j z_k z_l] = \delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}$, hence

$$\mathbb{E}[(z^\top \widetilde{M}z)(z^\top \widetilde{N}z)] = \sum_{i,j,k,l} \widetilde{M}_{ij}\widetilde{N}_{kl} \mathbb{E}[z_i z_j z_k z_l] = \text{tr} \widetilde{M} \text{tr} \widetilde{N} + 2 \text{tr}(\widetilde{M}\widetilde{N});$$

subtracting $\mathbb{E}[z^\top \widetilde{M}z] \mathbb{E}[z^\top \widetilde{N}z] = \text{tr} \widetilde{M} \text{tr} \widetilde{N}$ and unwhitening gives the result. $\square$

Lemma A.2 (Second-order expansion of Gaussian KL). *Let $u \mapsto \Sigma(u) \succ 0$ be twice continuously differentiable near 0 with $\Sigma(0) = \Sigma_0$ and differential $\dot{\Sigma}$ at 0. Then*

$$\text{KL}(\mathcal{N}(0, \Sigma_0) \parallel \mathcal{N}(0, \Sigma(u))) = \frac{1}{4} \text{tr} \left((\Sigma_0^{-1} \dot{\Sigma}[u])^2 \right) + o(\|u\|^2) = \frac{1}{2} G(u, u) + o(\|u\|^2),$$

the $\ddot{\Sigma}$ -terms cancelling identically.

Proof. $2 \text{KL} = \text{tr}(\Sigma(u)^{-1}\Sigma_0) - p + \log \det \Sigma(u) - \log \det \Sigma_0 =: g_1(u) + g_2(u) - p - \log \det \Sigma_0$. Along a ray $u = tv$, with $\Sigma = \Sigma(tv)$ and $\dot{\Sigma} = \frac{d}{dt}\Sigma$,

$$\dot{g}_1 = -\text{tr}(\Sigma^{-1}\dot{\Sigma}\Sigma^{-1}\Sigma_0), \quad \dot{g}_2 = \text{tr}(\Sigma^{-1}\dot{\Sigma}),$$

which at $t = 0$ are $-\text{tr}(\Sigma_0^{-1}\dot{\Sigma}[v])$ and $+\text{tr}(\Sigma_0^{-1}\dot{\Sigma}[v])$: they cancel, so 0 is a critical point. Differentiating once more and evaluating at $t = 0$ (using $\frac{d}{dt}\Sigma^{-1} = -\Sigma^{-1}\dot{\Sigma}\Sigma^{-1}$):

$$\ddot{g}_1(0) = 2\text{tr}((\Sigma_0^{-1}\dot{\Sigma}[v])^2) - \text{tr}(\Sigma_0^{-1}\ddot{\Sigma}[v, v]), \quad \ddot{g}_2(0) = \text{tr}(\Sigma_0^{-1}\ddot{\Sigma}[v, v]) - \text{tr}((\Sigma_0^{-1}\dot{\Sigma}[v])^2),$$

whose sum is $\text{tr}((\Sigma_0^{-1}\dot{\Sigma}[v])^2)$; hence $\text{KL}(tv) = \frac{t^2}{4}\text{tr}((\Sigma_0^{-1}\dot{\Sigma}[v])^2) + o(t^2)$. Since $u \mapsto \text{KL}$ is C^2 near 0, the ray-wise expansion with a quadratic form in v is a genuine second-order Taylor expansion, and $\frac{1}{4}\text{tr}((\Sigma_0^{-1}\dot{\Sigma}[u])^2) = \frac{1}{2}G(u, u)$ by (11). $\square$

B Linear-algebra toolbox

Lemma B.1 (Lyapunov operator). *For $S \succ 0$, $\mathbb{L}_S[X] = SX + XS$ is invertible on $\mathbb{R}^{q \times q}$, maps $\text{Sym}(q)$ onto $\text{Sym}(q)$ and $\text{Skew}(q)$ onto $\text{Skew}(q)$ bijectively, and $\|\mathbb{L}_S^{-1}[Y]\|_F \leq \|Y\|_F / (2\lambda_{\min}(S))$.*

Proof. In an orthonormal eigenbasis of S with eigenvalues $s_i > 0$, $(\mathbb{L}_S X)_{ij} = (s_i + s_j)X_{ij}$: entrywise multiplication by $s_i + s_j \geq 2\lambda_{\min}(S) > 0$. This is invertible with $(\mathbb{L}_S^{-1}Y)_{ij} = Y_{ij}/(s_i + s_j)$, preserves symmetry and skew-symmetry patterns (the multiplier is symmetric in (i, j)), and obeys the stated norm bound entrywise. $\square$

Lemma B.2 (Polar decomposition). *Every invertible $M \in \mathbb{R}^{q \times q}$ factors uniquely as $M = HR$ with $H \succ 0$ symmetric and $R \in \text{O}(q)$; explicitly $H = (MM^\top)^{1/2}$, $R = H^{-1}M$.*

Proof. Existence: with the SVD $M = \bar{U}\bar{S}\bar{V}^\top$ ($\bar{S} \succ 0$), take $H = \bar{U}\bar{S}\bar{U}^\top$ and $R = \bar{U}\bar{V}^\top$. Uniqueness: $M = HR$ forces $MM^\top = H^2$, and the positive-definite square root is unique, so H and then $R = H^{-1}M$ are determined. $\square$

Lemma B.3 (Orthogonal Procrustes). *Let $M := W'^\top W_0$ have SVD $M = \bar{U}\bar{S}\bar{V}^\top$. Then $\hat{R} = \bar{U}\bar{V}^\top$ minimizes $\|W'R - W_0\|_F$ over $R \in \text{O}(q)$; R is a minimizer iff $W_0^\top W'R$ is symmetric positive semidefinite; and the minimizer is unique iff M is nonsingular.*

Proof. $\|W'R - W_0\|_F^2 = \|W'\|_F^2 + \|W_0\|_F^2 - 2\text{tr}(R^\top M)$, so minimizing the distance is maximizing $\text{tr}(R^\top M)$. Writing $Z := \bar{U}^\top R\bar{V} \in \text{O}(q)$, $\text{tr}(R^\top M) = \text{tr}(R^\top \bar{U}\bar{S}\bar{V}^\top) = \text{tr}(Z^\top \bar{S}) = \sum_i S_{ii}Z_{ii} \leq \sum_i S_{ii} = \text{tr} \bar{S}$, since $|Z_{ii}| \leq 1$; the bound is attained by $Z = I$, i.e. $R = \bar{U}\bar{V}^\top$. If M is nonsingular then all $S_{ii} > 0$ and equality forces $Z_{ii} = 1$ for every i , which for an orthogonal Z forces $Z = I$ (unit diagonal entries of an orthogonal matrix force unit rows), giving uniqueness. Finally, at any maximizer, $W_0^\top W'R = M^\top R = \bar{V}\bar{S}\bar{U}^\top R = \bar{V}(Z^\top \bar{S})^\top \bar{V}^\top$ evaluated at $Z = I$ equals $\bar{V}\bar{S}\bar{V}^\top \succeq 0$; conversely if $W_0^\top W'R$ is symmetric $\succeq 0$ then $\text{tr}(R^\top M) = \text{tr}(W_0^\top W'R)$ equals the sum of the singular values of M (both matrices have the same singular values, and a symmetric psd matrix has trace equal to its singular-value sum), attaining the maximum. $\square$

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